

Corporate control under common ownership*

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Abstract

I show that the canonical *weighted average portfolio profit* model (WAPP) of firm conduct under common ownership imposes two conditions on firm conduct: (i) efficiency and (ii) independence of the distribution of power among shareholders from external factors such as the shareholders' stakes in competing firms, the competitors' strategies, and market conditions. If we drop the second condition, firm conduct can be represented through Nash bargaining (NB) among shareholders. I show how additional conditions on firm conduct characterize subclasses of WAPP and NB. The results have implications for robust testing of the common ownership hypothesis: When external factors influence the distribution of power among shareholders, tests assuming firm conduct is described by WAPP can fail to detect sizable common ownership effects of up to a 28.6% markup increase, a 26.5% consumer surplus loss, and an 8.2% total surplus loss compared to the own profit maximization benchmark.

Keywords: common ownership, corporate governance, institutional shareholders, bargaining, Nash bargaining, Nash-in-Nash, minority shareholdings, antitrust, competition policy, oligopoly

JEL classification codes: C71, D43, G34, L11, L13, L21, L41

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1 Introduction

Perfect competition is crucial for a firm’s shareholders to unanimously agree on own profit maximization (Hart, 1979), which has been the standard assumption on firm conduct since at least Fisher’s (1930) separation theorem. Yet, market power has been widespread and increasing in the U.S. economy and around the world (De Loecker and Eeckhout, 2018; De Loecker et al., 2020). At the same time, there is concern—and evidence—that the dramatic rise of common ownership (Backus et al., 2021a; Antón et al., 2025a) suppresses competition and reduces welfare by inducing firms to partially internalize their impact on competing firms’ profits (Rubinstein and Yaari, 1983; Rotemberg, 1984; Matvos and Ostrovsky, 2008; Elhauge, 2016; Posner et al., 2017; He and Huang, 2017; Azar et al., 2018; Schmalz, 2018; Azar and Vives, 2021; Lu et al., 2022; Li et al., 2023; Antón et al., 2023; Newham et al., 2025; Ederer and Pellegrino, 2025; Vives and Vravosinos, 2025; DOJ & FTC, 2023).¹ To be sure, such internalization can also have positive effects such as (i) increased innovation due to internalization of technological spillovers (López and Vives, 2019; Antón et al., 2025b) and (ii) prevention of R&D cost duplication across firms (Li et al., 2023; Vives and Vravosinos, 2025).

In modeling oligopolistic competition and studying the effects of common ownership—be it theoretically or empirically—a model of firm conduct other than own profit maximization is often necessary (see, e.g., Backus et al., 2021b; Ederer and Pellegrino, 2025; Vives and Vravosinos, 2025). The model needs to describe how the firm’s conduct is shaped by its shareholders’ conflicting interests. For example, shareholders with few shares of competing firms want the firm to price more aggressively than shareholders with larger stakes in competitors. Nevertheless, multiple authors have acknowledged our limited understanding around the modeling of firm conduct under common ownership (see, e.g., Schmalz, 2018; Backus et al., 2021a; Antón et al., 2023).

Most of the literature has so far relied on what I call the *weighted average portfolio profit* model (WAPP) of O’Brien and Salop (2000). Given a set $N := \{1, 2, \dots, n\}$ of shareholders and a set M of firms, let s_{if} denote shareholder i ’s number of firm $f \in M$ shares with cash-flow rights and π_g denote firm g ’s profit. WAPP posits that for every shareholder i , there exists control weight $\gamma_{if}(s_{*f})$ —which depends on the ownership structure $s_{*f} \equiv (s_{1f}, \dots, s_{nf})$ of firm f —such that firm f ’s conduct can be modeled as

¹Common ownership refers to the situation where competing firms have to an extent common shareholders. It has been increasing due to the expansion of not only public but also private equity funds (Li et al., 2023; Eldar and Grennan, 2024). While several studies find that common ownership affects firm conduct, there are also some that find little evidence to that effect (Backus et al., 2021b; Koch et al., 2021; Lewellen and Lowry, 2021). For a review of the common ownership literature, see Gerardi et al. (2024).

maximizing a weighted average of its shareholders' portfolio profits, that is,²

$$\sum_{i \in N} \overbrace{\gamma_{if}(s_{*f})}^{\text{shareholder } i\text{'s control weight over firm } f} \times \sum_{g \in M} \overbrace{s_{ig}\pi_g}^{\text{shareholder } i\text{'s portfolio profit}} \propto \pi_f + \sum_{g \in M \setminus \{f\}} \overbrace{\frac{\sum_{i \in N} \gamma_{if}(s_{*f}) s_{ig}}{\sum_{i \in N} \gamma_{if}(s_{*f}) s_{if}}}^{=: \lambda_{fg}(s)} \pi_g.$$

Equivalently, firm f maximizes its own profit plus each other firm g 's profit weighted by $\lambda_{fg}(s)$, the Edgeworth coefficient of effective sympathy of firm f towards firm g . The literature usually makes the *proportional control* assumption that $\gamma_{if}(s_{*f}) = s_{if}$.

Although WAPP is simple and instinctively reasonable, it imposes restrictions on firm conduct that merit careful study. Much of the discussion has revolved around the “correct” mapping γ_{*f} from ownership structure to control weights. Indeed, “any formulation of γ is implicitly a model of corporate governance and one where theory offers precious little guidance” (Backus et al., 2021a). The assumption that there even *exists* a correct mapping γ_{*f} such that the firm's conduct can be modeled as maximizing a weighted average of the shareholders' portfolio profits is considered less controversial (Backus et al., 2020). Nonetheless, it is not clear what restrictions it imposes on firm conduct. Therefore, a theoretical analysis is useful for (i) studying how properties of firm conduct translate into restrictions on the mapping γ_{*f} , (ii) evaluating the properties of firm conduct that allow it to even admit a WAPP representation, (iii) developing an alternative model of firm conduct under common ownership for when there is concern that firm conduct may not admit a WAPP representation, and (iv) assessing the implications of assuming firm conduct admits a WAPP representation when in reality it does not.

I pursue each of the four objectives by studying properties of the firm's *strategic plan*. The strategic plan describes the strategy the firm will follow as a function of the strategies followed by the other firms, the stakes of each shareholder across firms, and market conditions (e.g., demand or production technology). For example, WAPP with proportional control is a strategic plan. Clearly, one cannot hope to theoretically—or, for that matter, empirically—show that a specific strategic plan is the “correct” one if for no other reason than because different firms may have different strategic plans. Indeed, there are many reasonable strategic plans. Even if we constrain attention to WAPP, empirically estimating γ 's from data on market outcomes is a futile exercise given that there are usually many more shareholders than firms in an industry (Backus et al., 2020). Therefore, the focus of this paper is to characterize *classes* of strategic plans and subclasses thereof. For example, a class of strategic plans are those that admit a WAPP representation.

First, I propose two monotonicity properties that formally characterize the intuitive

²This formulation of WAPP is very general, encompassing arguably unrealistic models (e.g., one that specifies $\gamma_{if}(s_{*f}) = 1$ and $\gamma_{jf}(s_{*f}) = 0$ for some s_{*f} with $s_{if} = 1\%$ and $s_{jf} = 90\%$). Later on, I discuss how properties of firm conduct (e.g., capturing the notion that “more shares should lead to more control”) translate into restrictions on the mapping γ_{*f} .

notion that “more shares should lead to more control.” The first property, called *rank preservation*, relates to how the firm adjusts its strategy in response to changes in its shareholders’ interests. Starting from an industry ownership matrix $s \equiv (s_{if})_{i \in N, f \in M}$ where firm f shareholders’ interests are aligned, consider a stock trade between two shareholders i and j of firm f in which i buys shares of another firm $g \neq f$ from j . Before the stock trade, the shareholders’ interests are aligned. The stock trade causes disagreement among firm f shareholders: Shareholder i wants firm f to adjust its strategy in the direction benefiting firm g , while shareholder j wants firm f to adjust its strategy in the opposite direction. Firm f ’s strategic plan is rank-preserving if, in response to the stock trade, firm f adjusts its strategy in the direction preferred by the larger shareholder involved in the trade. I show that rank preservation is satisfied by WAPP if and only if $s_{if} \geq s_{jf} \implies \gamma_{if}(s_{*f}) \geq \gamma_{jf}(s_{*f})$. The second monotonicity property, *stock-trade monotonicity*, requires that a shareholder i ’s control power over firm f —as measured through firm f ’s strategy adjustment in response to stock trades involving shareholder i —increases when i grows her stake in every firm, including f . I show that stock-trade monotonicity is satisfied by WAPP if and only if $\gamma_{if}(s_{*f} + t(\mathbf{e}_i - \mathbf{e}_j)) \geq \gamma_{if}(s_{*f})$ for any $t > 0$, where $\mathbf{e}_i$ is the standard basis vector with 1 in its i -th entry. This implies γ_{if} (weakly) increases if shareholder i increases her stake in firm f while *no other* shareholder increases her stake in firm f . However, it does *not* imply that $s'_{if} \geq s_{if} \implies \gamma_{if}(s'_{*f}) \geq \gamma_{if}(s_{*f})$, which would be too strong (e.g., it would likely be violated for s_{*f} and s'_{*f} such that $s'_{jf} > 1/2 > s'_{if} \geq s_{if} = \max_{k \in N} s_{kf}$ for some $j \neq i$).

I also characterize a popular generalization of WAPP with proportional control positing that there exists a real function δ such that $\gamma_{if}(s_{*f}) = \delta(s_{if}) / \sum_{j \in N} \delta(s_{jf})$ (see, e.g., Backus et al., 2021a; Antón et al., 2023; Ederer and Pellegrino, 2025). Under proportional control, $\delta(s_{if}) = s_{if}$. I show that the control weights of a strategic plan that admits a WAPP representation can be expressed in this way if and only if the strategic plan satisfies three conditions: (i) anonymity, which requires that the identity of shareholders not matter for firm strategy, (ii) inclusivity, which requires that *every* shareholder of the firm—no matter how large or small—exerts some control over the firm, and (iii) independence of irrelevant shareholders (IIS), which requires that the relative control power of two shareholders over the firm be unaffected by stock trades between other shareholders of the firm.

Second, I show that two properties are crucial for the firm’s strategic plan to admit a WAPP representation: *efficiency* and *irrelevance of external factors*. Efficiency requires that for any ownership structure of the firm, there is a subset of controlling shareholders whose interests are efficiently pursued. Namely, the firm’s strategic plan always prescribes a strategy that is Pareto efficient with respect to the portfolio returns of the controlling shareholders. Irrelevance of external factors requires that the distribution of power among shareholders depend only on the firm’s ownership structure, and not on external factors such as (i) the stakes of firm f shareholders in competing firms, (ii) the strategies of

competitors, and (iii) market conditions.

Nevertheless, it is plausible that the distribution of power among a firm's shareholders can depend on such external factors. The dependence may arise because the distribution of power is shaped not only by the capacity of shareholders to exercise control but also by their *incentives* to do so. Firm f shareholders with larger stakes in competing firms may pay closer attention to the industry and thus exert more control over firm f (than other shareholders with the same number of firm f shares but fewer shares of competing firms). Indeed, there is evidence pointing in this direction (Van Nieuwerburgh and Veldkamp, 2010; Fich et al., 2015; Iliev et al., 2021). Also, firm f shareholders who also have shares of competing firm g may have stronger incentives to exert control over firm f when firm g 's strategy is such that firm f has a lot of room to influence g 's profitability (than when firm g 's strategy leaves little room for firm f to affect g 's profitability). For instance, if firm g expands (resp. limits) its presence and production capacity, firm f 's effect through its pricing and production strategy on firm g 's profit margin will have a significant (resp. modest) impact on firm g 's profits. Last, firm f shareholders who also have shares of competing firm g may have stronger incentives to try to affect firm f 's strategy when firm f and g 's goods are strong substitutes (than when they are weaker substitutes), in which case firm f 's pricing strategy will have a more pronounced impact on firm g 's profits.

I also show that the irrelevance of external factors condition rules out some natural benchmark strategic plans. Namely, it rules out the strategic plan that follows the median shareholder's interests, which can arise from majority voting (e.g., as in Chiappinelli et al., 2023). Firm f shareholders' preferences over firm f 's strategy depend not only on their stakes in firm f but also on (i) their stakes in competing firms, (ii) the strategies of competitors, and (iii) market conditions. Therefore, the identity of the median shareholder—and thus, the distribution of power among firm f shareholders—depends on all those factors. Thus, the strategic plan following the median shareholder's interests does *not* admit a WAPP representation. However, it does admit a Nash bargaining representation.

Third, I show that when external factors influence the distribution of power among shareholders, firm conduct can be represented as the result of asymmetric Nash bargaining (NB) among the firm's shareholders. The NB model generalizes WAPP by allowing for external factors to influence the distribution of power among shareholders while still requiring efficiency. The equilibrium concept is then a Nash equilibrium in Nash bargains, or simply Nash-in-Nash equilibrium. As in WAPP, I study the constraints imposed on NB by the monotonicity properties. I show that restrictions imposed on the bargaining weights based on intuition may fail to capture the notion that “more shares should lead to more control.” For example, one might expect that assigning larger shareholders higher bargaining weights in the Nash product would capture this notion. However, this is not the case. I show that a stronger condition characterizes rank-preserving NB strategic plans.

Fourth, I study the implications of firm conduct model misspecification. I show that when external factors influence the distribution of power among shareholders—and, thus, firm conduct admits an NB but *not* WAPP representation—own profit maximization may explain firm conduct better than *every* rank-preserving specification of WAPP, even though common ownership affects firm conduct. Thus, failing to account for how external factors might affect the distribution of power among shareholders can lead to a false rejection of the common ownership hypothesis (i.e., the hypothesis that common ownership induces firms to partially internalize their effect on competing firms’ profits). I show that this can lead to sizable effects of common ownership going undetected. Common ownership hypothesis tests that incorrectly assume firm conduct to admit a WAPP representation can fail to detect common ownership effects that lead to a markup increase of up to 28.6%, consumer surplus loss of up to 26.5%, and total surplus loss of up to 8.2% relative to the benchmark where each firm maximizes its own profit.

The results contribute to our understanding around the modeling of firm conduct under common ownership. They can help researchers and practitioners carefully evaluate the assumptions imposed on firm conduct by the widely used WAPP model, and by the popular generalization of WAPP with proportional control. The results can guide empirical researchers and policymakers on how to robustly test the common ownership hypothesis when there is concern that the irrelevance of external factors condition imposed by WAPP might be violated.

Last, I show that the NB model can relax the tension that arises under WAPP between (i) allowing for atomistic shareholders to collectively exert control over the firm and (ii) allowing for large shareholders to have control power. As has been noted before (see, e.g., Gramlich and Grundl, 2017; O’Brien and Waehrer, 2017; Brito et al., 2023), I show that generally, under WAPP, as ownership by a group of shareholders with aligned interests is diffused (i.e., the same number of shares across firms is divided across more and more shareholders), the group of shareholders stops having any control over firm strategy. For example, as shareholders who do not hold shares of competing firms become dispersed, the firm tends to follow only the interests of the shareholders who are also invested in competitors. While this may be plausible, the possibility that atomistic shareholders collectively exert control over the firm is also plausible. Nonetheless, most parametrizations of the WAPP model preclude this possibility. I show that those parametrizations that do allow for atomistic shareholders to collectively exert control over the firm have an unrealistic property: They assign no control power to large shareholders when atomistic shareholders are also present. NB allows for the possibility that atomistic shareholders collectively exert control over the firm without imposing this unrealistic property.

After a discussion of related literature, section 2 defines WAPP and NB strategic plans. Section 3 characterizes WAPP and NB and studies how properties of firm conduct translate into restrictions on the parameters of the firm’s objective under WAPP and

NB. Section 4 discusses the properties of firm conduct studied in section 3, as well as (i) the effects of ownership dispersion on firm conduct, (ii) potential formulations of the disagreement payoffs in NB, and (iii) the assumptions imposed in some of the analysis of section 3. Section 5 studies the implications for common ownership hypothesis testing. Section 6 concludes. All proofs are gathered in the Appendix. The Online Appendix provides supplementary results.

Related literature O’Brien and Salop (2000) proposed the WAPP model, while similar ideas can be traced back to Rotemberg (1984) and Bresnahan and Salop (1986). Since then, the limited work studying how to model firm conduct under common ownership has primarily focused on microfounding WAPP in models of shareholder voting (see, e.g., Azar, 2017; Brito et al., 2018; Moskalev, 2019). Brito et al. (2023) propose a voting model which unlike standard formulations of WAPP, allows for atomistic shareholders to collectively exert control over the firm. However, it prevents large shareholders from having control power. Indeed, in the Online Appendix, I show that Brito et al.’s (2023) model is WAPP, which is why it cannot allow for atomistic shareholders to collectively exert control over the firm while also allowing for large shareholders to have control power. On the other hand, Chiappinelli et al. (2023) consider a setting where shareholders elect managers through majority rule, and as a result the median shareholder’s preferences dictate firm strategy. I show that the resulting strategic plan is efficient but induces a distribution of power among shareholders that depends on external factors. Therefore, it admits an NB representation but not a WAPP one. Edmans et al. (2019) study how common ownership might strengthen governance through both voice and exit.

The contribution of the current paper is to characterize the whole WAPP class of firm conduct models, as well as the more general class of NB models. Also, I quantify the implications of restricting attention to WAPP in conduct testing. My approach differs methodologically from previous work. Instead of making an ad-hoc assumption on the firm’s objective function or microfounding a firm conduct model through shareholder voting, I follow an axiomatic approach. I discuss properties of firm conduct and show how these properties translate into formulations of the firm’s objective function.

If every firm’s strategic plan is NB, the equilibrium concept is Nash-in-Nash, which has become a standard tool since Horn and Wolinsky (1988) used it to study merger incentives when there are exclusive vertical relationships. The current paper fits into the literature that has leveraged the Nash-in-Nash solution to study equilibrium outcomes in a broad range of environments where the problem of the division of surplus between parties (e.g., upstream and downstream firms) is embedded within a larger market model. For a review of the Nash-in-Nash literature, see Collard-Wexler et al. (2019).

2 Strategic plans

A tuple $G := \langle N, M, (A_f)_{f \in M}, (\pi_f)_{f \in M}, (s_{if})_{(i,f) \in N \times M} \rangle$ characterizes an oligopoly game with common ownership, where $N := \{1, 2, \dots, n\}$ is the set of shareholders, $M := \{1, 2, \dots, m\}$ is the set of firms (in some industry or product market), and A_f is firm f 's strategy space. We will use i, j, k to denote shareholders and f, g, h , to denote firms. Let the strategy profile space be denoted by $A := \times_{f \in M} A_f$. For a strategy profile $a \equiv (a_1, \dots, a_m) \in A$, where $a_f \in A_f$ is firm f 's strategy, a_{-f} denotes the strategy profile of all firms except f , and accordingly $A_{-f} := \times_{g \neq f} A_g$. Firm f 's profit function is $\pi_f : A \rightarrow \mathbb{R}$, and $s \in S := \{s \in [0, 1]^{n \times m} : \sum_{i \in N} s_{if} = 1 \ \forall f \in M\}$ is the ownership matrix of the firms, where s_{if} is shareholder i 's share of firm f . s_{*f} is firm f 's ownership structure.³ This means that i has a cash-flow right over fraction s_{if} of firm f 's profits. Shareholder i 's total portfolio profit function is $u_i(a, s_{i*}) := \sum_{f \in M} s_{if} \pi_f(a)$. A shareholder i is a shareholder of firm f if $s_{if} > 0$. $N_f(s_{*f}) := \{i \in N : s_{if} > 0\}$ is the set of shareholders of firm f .

A strategic plan $R_f : \times_{g \neq f} \Delta(A_g) \times S \times \Pi \rightarrow \Delta(A_f)$ of firm f determines the nonempty set $R_f(\alpha_{-f}, s; \pi)$ of strategies deemed choosable by firm f for each strategy profile α_{-f} of the other firms, each ownership matrix s , and each vector $\pi \equiv (\pi_1, \dots, \pi_m) \in \Pi$ of profit functions, where $\Delta(A_g)$ is the set of lotteries over A_g with finite support and Π is the set of possible vectors of profit functions. The strategic plan describes firm conduct for any possible π given that market conditions such as technology and demand may change. To simplify notation, I write $u_i(a, s_{i*})$ instead of $u_i(a, s_{i*}; \pi)$. Abusing notation, we also write $u_i(\alpha, s_{i*})$ and $\pi_f(\alpha)$ to denote shareholder i 's expected portfolio profit and firm f 's expected profit, respectively.

Given an s and π , a strategy profile is an equilibrium if every firm plays one of its choosable strategies given the other firms' strategies.

2.1 The weighted average portfolio profit (WAPP) strategic plan

Let $\Delta^n \subset \mathbb{R}^n$ denote the $(n - 1)$ -dimensional simplex. I first describe the strategic plan proposed by O'Brien and Salop (2000), which I call the weighted average portfolio profit strategic plan (WAPP).

Definition 1. Firm f 's strategic plan R_f is WAPP if there exists a control power function $\gamma_{*f} : \Delta^n \rightarrow \Delta^n$ such that for every $s \in S$, $\alpha_{-f} \in \times_{g \neq f} \Delta(A_g)$, and $\pi \in \Pi$,

³As in the majority of the literature, s is treated as exogenous. Denicolò and Panunzi (2025) and Piccolo and Schneemeier (2025) endogenize common ownership. The analysis goes through if we extend the model to allow for short positions (where those shorting a firm's stock do not exert control over the firm). Given a matrix x , x_{i*} and x_{*f} denote its i -th row and f -th column, respectively.

(i) the firm maximizes the weighted average portfolio profit of its shareholders:

$$R_f(\alpha_{-f}, s; \pi) = \arg \max_{\alpha_f \in \Delta(A_f)} \left\{ \sum_{i \in N} \gamma_{if}(s_{*f}) u_i(\alpha_f, \alpha_{-f}, s_{i*}) \right\},$$

(ii) control is exclusive to shareholders: For every $i \in N$, $s_{if} = 0 \implies \gamma_{if}(s_{*f}) = 0$.

This can be rewritten as

$$R_f(\alpha_{-f}, s; \pi) = \arg \max_{\alpha_f} \left\{ \pi_f(\alpha_f, \alpha_{-f}) + \sum_{g \in M \setminus \{f\}} \overbrace{\frac{\sum_{i \in N} \gamma_{if}(s_{*f}) s_{ig}}{\sum_{i \in N} \gamma_{if}(s_{*f}) s_{if}}}^{=: \lambda_{fg}(s) \geq 0} \pi_g(\alpha_f, \alpha_{-f}) \right\},$$

where $\lambda_{fg}(s)$ is the weight firm f places on firm g 's profit with λ_{ff} normalized to 1. λ_{fg} is called the Edgeworth (1881) coefficient of effective sympathy of firm f towards firm g . The numerator of λ_{fg} is a measure of the level of cross-holdings of firm f shareholders in firm g . The denominator measures ownership concentration in firm f .

The literature often assumes $\gamma_{*f}(s_{*f}) = s_{*f}$, which it calls “proportional control.” A popular generalization of proportional control specifies $\gamma_{if}(s_{*f}) = \delta(s_{if}) / \sum_{j \in N} \delta(s_{jf})$ for some non-negative and non-decreasing real function δ . Particularly, $\delta(s_{if}) = s_{if}^\zeta$ for some $\zeta \geq 0$ is often used (see, e.g., Backus et al., 2021a; Antón et al., 2023; Ederer and Pellegrino, 2025). $\zeta > 1$ is interpreted as large shareholders having disproportionately more power than smaller shareholders. $\zeta = 1$ corresponds to proportional control. $\zeta < 1$ is interpreted as large shareholders having less than proportionately more power than smaller shareholders. Last, Azar et al. (2018), Brito et al. (2023), and Antón et al. (2025a) consider the case where γ_{if} is shareholder i 's (normalized) Banzhaf power index (Penrose, 1946; Banzhaf, 1965; Coleman, 1971). In that case, to calculate γ_{if} , one first enumerates all winning (*i.e.*, with at least 50% of the firm's shares) coalitions of shareholders where there is (at least) one swing shareholder (*i.e.*, a shareholder who is in the coalition and by leaving it would prevent the coalition from reaching majority). γ_{if} is the share of such coalitions where i is a swing shareholder, that is,⁴

$$\gamma_{if}(s_{*f}) = \frac{\left| \left\{ T \in 2^N : \sum_{j \in T} s_{jf} \geq 1/2 > \sum_{j \in T \setminus \{i\}} s_{jf} \right\} \right|}{\sum_{k \in N} \left| \left\{ T \in 2^N : \sum_{j \in T} s_{jf} \geq 1/2 > \sum_{j \in T \setminus \{k\}} s_{jf} \right\} \right|}.$$

2.2 The Nash bargaining (NB) strategic plan

I now describe the Nash bargaining strategic plan (NB).

Definition 2. Firm f 's strategic plan R_f is NB if there exist a bargaining power function $\beta_{*f} : \Delta^n \rightarrow \Delta^n$ and a disagreement payoff function $d_{*f} : \times_{g \neq f} \Delta(A_g) \times S \times \Pi \rightarrow \mathbb{R}^n$ such

⁴Each coalition is counted as many times as its number of swing shareholders.

that for every $s \in S$, $\alpha_{-f} \in \times_{g \neq f} \Delta(A_g)$, and $\pi \in \Pi$,

(i) the firm maximizes the Nash product:

$$R_f(\alpha_{-f}, s; \pi) = \arg \max_{\alpha_f \in B_f(\alpha_{-f}, s; \pi)} \left\{ \prod_{i \in N_f(\beta_{*f}(s_{*f}))} (u_i(\alpha_f, \alpha_{-f}, s_{i*}) - d_{if}(\alpha_{-f}, s; \pi))^{\beta_{if}(s_{*f})} \right\},$$

where $B_f(\alpha_{-f}, s; \pi) := \{\alpha_f \in \Delta(A_f) : u_i(\alpha_f, \alpha_{-f}, s_{i*}) \geq d_{if}(\alpha_{-f}, s; \pi) \forall i \in N_f(\beta_{*f}(s_{*f}))\}$
and $N_f(\beta_{*f}(s_{*f})) \equiv \{i \in N : \beta_{if}(s_{*f}) > 0\}$,

(ii) control is exclusive to shareholders: For every $i \in N$, $s_{if} = 0 \implies \beta_{if}(s_{*f}) = 0$.

If the maximum Nash product is positive for every s , π , and α_{-f} , we say that there are strict benefits from agreement.

Remark: Given that by definition, R_f is nonempty-valued, $B_f(\alpha_{-f}, s; \pi)$ has to be nonempty.

The analysis goes through if we require that there exist function $\alpha_f^d : \times_{g \neq f} \Delta(A_g) \times S \times \Pi \rightarrow \Delta(A_f)$ such that for every i , α_{-f} , s , and π , $d_{if}(\alpha_{-f}, s; \pi) = u_i(\alpha_f^d(\alpha_{-f}, s; \pi), \alpha_{-f}, s_{i*})$, where $\alpha_f^d(\alpha_{-f}, s; \pi)$ is the strategy chosen by firm f in case of disagreement when the other firms' strategies are α_{-f} , the ownership matrix is s , and the profit functions are π . It is reasonable for firm f 's strategy in case of disagreement to depend on α_{-f} , s , and π . For example, if firm f 's competitors produce large quantities or set low prices driving f 's residual demand down, $\alpha_f^d(\alpha_{-f}, s; \pi)$ will probably reflect that f should not produce a lot in case of disagreement. Section 4.6 discusses potential formulations of $\alpha_f^d(\alpha_{-f}, s; \pi)$.

When each firm's strategic plan is NB, the equilibrium concept is Nash-in-Nash. While the focus of the paper is a single firm's strategic plan, the Online Appendix provides sufficient conditions for the existence of a Nash-in-Nash equilibrium. It also uses Nash-in-Nash to study the competitive effects of changes in corporate control providing a rationale for a policy proposal by Posner et al. (2017) requiring institutional investors to be passive.

3 Properties of strategic plans

This section discusses properties of strategic plans. First, it shows that assuming a strategic plan is NB is almost equivalent to requiring that the strategic plan (i) satisfy a form of Pareto efficiency and (ii) prescribes strategies that are unique up to payoff-equivalent strategies. Assuming that a strategic plan is WAPP also implies that the strategic plan must satisfy this efficiency condition but it also imposes an additional restriction: The distribution of power among shareholders must be independent of external factors such as (i) the other firms' ownership structures, (ii) the other firms' strategies, and (iii) market conditions (e.g., market demand or production technology). Second, it proposes two monotonicity properties capturing the notion that "more shares should lead to more

control” and characterizes when WAPP and NB strategic plans satisfy those properties. Third, it characterizes a class of WAPP strategic plans where $\gamma_{if}(s_{*f}) = \delta(s_{if}) / \sum_{j \in N} \delta(s_{jf})$ for some real function δ , which generalizes WAPP with proportional control.

3.1 Efficiency and internal consistency

Firm f ’s strategic plan is efficient if under any ownership structure s_{*f} of the firm, there is a subset $\widetilde{N}(s_{*f})$ of the shareholders of firm f who efficiently control the firm. Strong efficiency requires that for any strategy profile of the other firms, firm f never responds with a weakly Pareto dominated strategy—in the sense that another strategy could do at least as well for all controlling shareholders and strictly better for at least one of them. Weak efficiency requires that for any strategy profile of the other firms, firm f never responds with a strongly Pareto dominated strategy.

Definition 3. The strategic plan R_f of firm f is strongly (resp. weakly) efficient if there exists correspondence $\widetilde{N} : \Delta^n \rightrightarrows N$ such that for every $s \in S$,

- (i) a nonempty set $\widetilde{N}(s_{*f})$ of shareholders control the firm,
- (ii) control is exclusive to shareholders: For every $i \in N$, $s_{if} = 0 \implies i \notin \widetilde{N}(s_{*f})$, and
- (iii) the firm is efficiently controlled: For every $\pi \in \Pi$ and $\alpha_{-f} \in \times_{g \neq f} \Delta(A_g)$, there do not exist $\alpha'_f \in \Delta(A_f)$ and $\alpha_f \in R_f(\alpha_{-f}, s; \pi)$ such that $u_i(\alpha'_f, \alpha_{-f}, s_{i*}) \geq u_i(\alpha_f, \alpha_{-f}, s_{i*})$ for all $i \in \widetilde{N}(s_{*f})$ with at least one (resp. every) inequality strict.

Firm f ’s strategic plan R_f is internally consistent if, in addition, for every $\pi \in \Pi$, $s \in S$, $\alpha_{-f} \in \times_{g \neq f} \Delta(A_g)$, $\alpha_f, \alpha'_f \in R_f(\alpha_{-f}, s; \pi)$, and $\alpha''_f \in \Delta(A_f)$

- (iv) $u_i(\alpha_f, \alpha_{-f}, s_{i*}) = u_i(\alpha'_f, \alpha_{-f}, s_{i*})$ for all $i \in \widetilde{N}(s_{*f})$, and
- (v) if $u_i(\alpha_f, \alpha_{-f}, s_{i*}) = u_i(\alpha''_f, \alpha_{-f}, s_{i*})$ for all $i \in \widetilde{N}(s_{*f})$, then $\alpha''_f \in R_f(\alpha_{-f}, s; \pi)$.

Remark: This definition of efficiency implicitly assumes that there are frictions preventing monetary transfers between shareholders that could contribute to Pareto improvements. For a related discussion, see section 4.5.

Internal consistency requires that the firm’s strategic plan prescribes strategies that are unique up to payoff-equivalent strategies. With Pareto-dominated strategies already ruled out by efficiency, condition (iv) requires that firm f ’s controlling shareholders not be willing to agree to two different strategies α_f and α'_f when α_f is strictly preferred to α'_f by one controlling shareholder and α'_f is strictly preferred to α_f by another shareholder. Also, if firm f ’s controlling shareholders are willing to agree to strategy α_f , condition (v) requires that they also be willing to agree to any other strategy that delivers the same portfolio profit to each one of them.

Let $\mathcal{U}_f(\alpha_{-f}, s; \pi) := \{v \in \mathbb{R}^{|N_f(s_{*f})|} : \exists \alpha_f \in \Delta(A_f) \text{ such that } u_i(\alpha_f, \alpha_{-f}, s_{i*}) = v_i \text{ for every } i \in N_f(s_{*f})\}$ denote the (convex) portfolio profit possibility set of firm f shareholders when the other firms' strategies are α_{-f} , the ownership matrix is s , and the profit functions are π . Proposition 1 studies the efficiency and internal consistency properties of WAPP and NB strategic plans.

Proposition 1. Consider a firm $f \in M$.

- (i) If R_f is WAPP, then it is strongly efficient.
- (ii) If R_f is NB, then it is weakly efficient.
- (iii) If R_f is NB with strict benefits from agreement, then it is strongly efficient and internally consistent.
- (iv) If R_f is strongly efficient and internally consistent, then it is NB.
- (v) Assume A_f is convex and that for every $s \in S$, $a_{-f} \in A_{-f}$, $\pi \in \Pi$, and $i \in N_f(s_{*f})$, $u_i(a_f, a_{-f}, s_{i*})$ is strictly concave in a_f . If R_f is WAPP or NB, then it is strongly efficient and internally consistent.

Remark: Lemma 3 in the Online Appendix gives sufficient conditions for the assumption in part (v) to hold in a homogeneous product Cournot market.

Parts (ii)-(iv) show that assuming a strategic plan is NB is approximately equivalent to assuming it is efficient and internally consistent. Under the condition in part (v), the class of WAPP strategic plans is a subset of the class of NB strategic plans, which coincides with the class of strongly efficient and internally consistent strategic plans. Without that condition, from parts (i) and (iv), it follows that the class of WAPP and internally consistent strategic plans is a subset of the class of NB strategic plans. The only case where R_f can be WAPP but not NB is when that condition fails and R_f is *not* internally consistent. In that case, a WAPP R_f can choose all strategies that result in portfolio profit profiles of firm f 's controlling shareholders across a segment of $\mathcal{U}_f(\alpha_{-f}, s; \pi)$ along which the WAPP objective is constant, which an NB R_f cannot do. Therefore, barring a small and arguably uninteresting class of internally inconsistent WAPP strategic plans, the class of WAPP strategic plans is a subset of the class of NB strategic plans.

3.2 (Ir)relevance of external factors

Any model of firm conduct must capture the fact that the distribution of power among a firm's shareholders depends on factors *internal* to firm f . Particularly, the number of shares held by each shareholder can play an important role: Larger shareholders can be expected to have greater power in shaping firm conduct than smaller ones. Indeed, NB and WAPP can capture this given that γ_{*f} and β_{*f} depend on s_{*f} .

At the same time, the distribution of power among firm f shareholders may depend also on *external* factors such as (i) the stakes of firm f shareholders in competing firms, (ii) the strategies of other firms, and (iii) market conditions (e.g., demand or production technology). WAPP precludes such dependence of the distribution $\gamma_{*f}(s_{*f})$ of power across a firm's shareholders on *external* factors, while NB allows for it. To see this, let A_f be a subset of a Euclidean space with $R_f(\alpha_{-f}, s; \pi)$ pinned down by the first-order condition (FOC). The FOC under WAPP is $\sum_{i \in N} \gamma_{if}(s_{*f}) \partial u_i(a_f, \alpha_{-f}, s_{i*}) / \partial a_f|_{a_f=R_f(\alpha_{-f}, s; \pi)} = \mathbf{0}$, where $\partial u_i(a_f, \alpha_{-f}, s_{i*}) / \partial a_f$ is the gradient with respect to a_f . Under NB, the FOC is $\sum_{i \in N_f(\beta_{*f}(s_{*f}))} \tilde{\gamma}_{if}(\alpha_{-f}, s; \pi) \partial u_i(a_f, \alpha_{-f}, s_{i*}) / \partial a_f|_{a_f=R_f(\alpha_{-f}, s; \pi)} = \mathbf{0}$, where

$$\tilde{\gamma}_{if}(\alpha_{-f}, s; \pi) := \frac{\frac{\beta_{if}(s_{*f})}{u_i(R_f(\alpha_{-f}, s; \pi), \alpha_{-f}, s_{i*}) - d_{if}(\alpha_{-f}, s; \pi)}}{\sum_{j \in N_f(\beta_{*f}(s_{*f}))} \frac{\beta_{jf}(s_{*f})}{u_j(R_f(\alpha_{-f}, s; \pi), \alpha_{-f}, s_{j*}) - d_{jf}(\alpha_{-f}, s; \pi)}}$$

is the disagreement-adjusted control power of shareholder i over firm f at $(\alpha_{-f}, s; \pi)$. It measures shareholder control accounting for the fact that the further u_i is from d_{if} , the more shareholder i has to lose in case of disagreement. NB allows for $\tilde{\gamma}_{*f}(\alpha_{-f}, s; \pi)$ to depend on the other firms' strategies or ownership structures and on market conditions, since disagreement payoffs depend on those.⁵ Equivalently, we can write

$$\left. \frac{\partial \pi_f(a_f, \alpha_{-f})}{\partial a_f} \right|_{a_f=R_f(\alpha_{-f}, s; \pi)} + \sum_{g \in M \setminus \{f\}} \tilde{\lambda}_{fg}(\alpha_{-f}, s; \pi) \left. \frac{\partial \pi_g(a_f, \alpha_{-f})}{\partial a_f} \right|_{a_f=R_f(\alpha_{-f}, s; \pi)} = \mathbf{0},$$

where $\tilde{\lambda}_{fg}(\alpha_{-f}, s; \pi) := \sum_{i \in N_f(\beta_{*f}(s_{*f}))} \tilde{\gamma}_{if}(\alpha_{-f}, s; \pi) s_{ig} / \sum_{i \in N_f(\beta_{*f}(s_{*f}))} \tilde{\gamma}_{if}(\alpha_{-f}, s; \pi) s_{if}$ is the weight firm f places on firm g 's profit.

In fact, the irrelevance of external factors for the distribution of power among shareholders is what characterizes WAPP as a special case of NB. To see this, define a generalized weighted average portfolio profit strategic plan (GWAPP) as follows.

Definition 4. Firm f 's strategic plan R_f is GWAPP if there exists a control power function $\gamma_{*f} : \times_{g \neq f} \Delta(A_g) \times S \times \Pi \rightarrow \Delta^n$ such that for every $s \in S$, $\alpha_{-f} \in \times_{g \neq f} \Delta(A_g)$, and $\pi \in \Pi$,

- (i) the firm maximizes the weighted average portfolio profit of its shareholders:

$$R_f(\alpha_{-f}, s; \pi) = \arg \max_{\alpha_f \in \Delta(A_f)} \left\{ \sum_{i \in N} \gamma_{if}(\alpha_{-f}, s; \pi) u_i(\alpha_f, \alpha_{-f}, s_{i*}) \right\},$$

- (ii) control is exclusive to shareholders: For every $i \in N$, $s_{if} = 0 \implies \gamma_{if}(\alpha_{-f}, s; \pi) = 0$,

⁵This is the case even when the strategy $\alpha_f^d(\alpha_{-f}, s; \pi)$ chosen by firm f in case of disagreement (if disagreement payoffs are determined by such strategy) is independent of α_{-f} , s , and π .

- (iii) external factors can influence a shareholder's magnitude of control power but not whether she has control power or not: For every $s' \in S$, $\alpha'_{-f} \in \times_{g \neq f} \Delta(A_g)$, and $\pi' \in \Pi$ such that $s_{*f} = s'_{*f}$, $N_f(\gamma_{*f}(\alpha_{-f}, s; \pi)) \equiv \{i \in N : \gamma_{if}(\alpha_{-f}, s; \pi) > 0\} = N_f(\gamma_{*f}(\alpha'_{-f}, s'; \pi'))$.

Remark: If R_f is WAPP, it is also GWAPP. The third point mirrors the following observation about NB: Although NB allows for the distribution of power among shareholders to depend on external factors, the set of firm f controlling shareholders can depend only on f 's ownership structure (see section 3.1).

Proposition 2 shows that under the condition in part (v) of Proposition 1, the class of NB strategic plans with strict benefits from agreement is a subset of the class of GWAPP strategic plans, which in turn is a subset of the class of NB strategic plans. Without that condition, from part (i) of Proposition 2 and part (iv) of Proposition 1, it follows that, barring a class of internally inconsistent GWAPP strategic plans, the class of GWAPP strategic plans is a subset of the class of NB strategic plans.

Proposition 2. Consider a firm $f \in M$.

- (i) If R_f is GWAPP, then it is strongly efficient.
- (ii) Assume the condition of part (v) of Proposition 1 holds. Then, (a) if R_f is GWAPP, it is NB; and (b) if R_f is NB with strict benefits from agreement, it is GWAPP.

An advantage of the NB formulation over GWAPP is that it can be more natural to specify firm f 's strategy in case of disagreement—and thereby indirectly specify how s , α_{-f} , and π affect γ_{*f} . Within the GWAPP framework, it is hard to directly specify the mapping from α_{-f} , s , and π to γ_{*f} . Restricting attention to WAPP strategic plans restricts attention to GWAPP strategic plans with $\gamma_{*f}(\alpha_{-f}, s; \pi)$ independent of α_{-f} , π , and $(s_{*g})_{g \neq f}$.⁶

3.3 Monotonicity

In this section, I study monotonicity properties of firm f 's strategic plan. The properties impose conditions on how the strategy prescribed by the strategic plan changes when firm f shareholders trade shares of some other firm $g \neq f$. Before proceeding, we need to develop a language to talk about stock trades. For every shareholder $i \in N_f(s_{*f})$ of firm f , $(\lambda_{i;f1}, \lambda_{i;f2}, \dots, \lambda_{i;fm}) \equiv \lambda_{i;f*} := \frac{1}{s_{if}} s_{i*} \equiv (s_{i1}/s_{if}, s_{i2}/s_{if}, \dots, s_{im}/s_{if})$, is the vector of weights i wants firm f to place on firms' profits with the weight on firm f 's profit normalized to 1.

⁶Lemma 1 in section 3.3 characterizes what γ_{*f} captures. It is possible to characterize the independence of $\gamma_{*f}(\alpha_{-f}, s; \pi)$ from α_{-f} , π , and $(s_{*g})_{g \neq f}$ using f -neutral stock trades as in Lemma 1 and Proposition 4. However, the main intuition is provided by Lemma 1 itself.

Definition 5. Firm f shareholders unanimously agree on firm conduct under ownership matrix s if $\lambda_{i,f*} = \lambda_{j,f*}$ for every $i, j \in N_f(s_{*f})$. Then, s is called f -unanimous.

For simplicity, the remainder of section 3, let A_f be an interval and restrict attention to pure strategies. Also, fix some vector of profit functions $\pi \in \Pi$ such that for every f -unanimous s and $g \neq f$, there exists $a_{-f} \in A_{-f}$ such that $\partial \pi_g(a_f, a_{-f}) / \partial a_f \neq 0$ evaluated at $a_f = R_f(a_{-f}, s; \pi)$. This assumption requires at least a minimal level of externalities from firm f on other firms. Without such externalities, firm f shareholders' preferences over firm f 's conduct would not depend on their stakes in other firms. To economize on notation, we suppress the dependence of all objects on π (e.g., by writing $R_f(a_{-f}, s)$ instead of $R_f(a_{-f}, s; \pi)$). Whether R_f is WAPP or NB, assume that for every a_{-f} and f -unanimous s , the firm's objective is continuously differentiable with $R_f(a_{-f}, s)$ pinned down by the FOC and the second-order condition holding strictly.⁷ Also, assume that under NB, there exists function $\alpha_f^d : \times \Delta(A_g)_{g \neq f} \times S \rightarrow \Delta(A_f)$ such that for every a_{-f} , s , and $i \in N$, $d_{if}(a_{-f}, s) = u_i(\alpha_f^d(a_{-f}, s), a_{-f}, s_{i*})$, where $\alpha_f^d(a_{-f}, s)$ is the strategy chosen by firm f in case of disagreement.

We will see that studying the firm's strategic plan locally, around f -unanimous matrices, is a powerful approach. Starting from an f -unanimous matrix, a small stock trade where some firm f shareholders trade firm $g \neq f$ shares can cause firm f to adjust its strategy only through its effect on firm f shareholders' interests. Even if the stock trade affects the distribution of power among firm f shareholders (which is possible if firm f 's strategic plan is NB), this will not play a role in how firm f adjusts its strategy in response to the stock trade: Given that firm f shareholders unanimously agree on firm strategy to begin with, changes in the distribution of power among them are inconsequential.

Definition 6. A $(\psi, g, i, \widetilde{N})$ -stock trade is an infinitesimal change in the ownership matrix s in direction $ds = [(1 - \psi)\mathbf{e}_i - \psi \sum_{j \in \widetilde{N}} \mathbf{e}_j] \otimes \mathbf{e}_g$, where $\psi \in [0, 1]$, $g \in M$, $i \in N \setminus \widetilde{N}$, $\emptyset \neq \widetilde{N} \subset N$, and $\otimes$ denotes the outer product.

In a $(\psi, g, i, \widetilde{N})$ -stock trade, shareholder i buys firm g shares at rate $1 - \psi$, and each shareholder in group $\widetilde{N}$ of shareholders sells firm g shares at rate ψ . We will study how firm f adjusts its strategy in response to a $(\psi, g, i, \widetilde{N})$ -stock trade where $g \neq f$, $i \in N_f(s_{*f})$, and $\widetilde{N} \subset N_f(s_{*f})$.

Strictly put, a $(\psi, g, i, \widetilde{N})$ -stock trade is possible only when $\psi|\widetilde{N}| = 1 - \psi$, or equivalently, $\psi = (|\widetilde{N}| + 1)^{-1}$, so that ds points inside S . $\psi \neq (|\widetilde{N}| + 1)^{-1}$ can be interpreted as follows: There are some additional firm g shareholders outside the set N without control power over firm f . When $\psi < (|\widetilde{N}| + 1)^{-1}$, these additional shareholders sell firm g shares to

⁷That is, under WAPP, $\sum_{i \in N} \gamma_{if}(s_{*f}) \partial u_i(a_f, a_{-f}, s_{i*}) / \partial a_f|_{a_f=R_f(a_{-f}, s)}$ is continuously differentiable in (a_f, s) . Under NB, $\sum_{i \in N_f(\beta_{*f}(s_{*f}))} \tilde{\gamma}_{if}(a_{-f}, s) \partial u_i(a_f, a_{-f}, s_{i*}) / \partial a_f|_{a_f=R_f(a_{-f}, s; \pi)}$ is continuously differentiable in (a_f, s) .

shareholder i at rate $1 - \psi(|\widetilde{N}| + 1)$. When $\psi > (|\widetilde{N}| + 1)^{-1}$, these additional shareholders buy firm g shares from group $\widetilde{N}$ of shareholders at rate $\psi(|\widetilde{N}| + 1) - 1$.⁸

3.3.1 Rank preservation

When $\widetilde{N} = \{j\}$ is a singleton, we call the stock trade a (ψ, g, i, j) -stock trade. In a $(1/2, g, i, j)$ -stock trade, shareholder i buys firm g shares from shareholder j . We are now ready to study the first monotonicity property: *rank preservation*.

Definition 7. Firm f 's strategic plan is rank-preserving if for any firm $g \neq f$, any strategy profile a_{-f} of the other firms, any f -unanimous ownership matrix s , and any pair of distinct shareholders $i, j \in N_f(s_{*f})$, if $s_{if} \geq s_{jf}$, then the directional derivative $\nabla_{ds} R_f(a_{-f}, s)$ given a $(1/2, g, i, j)$ -stock trade (which changes the ownership matrix in direction ds) satisfies

$$\left. \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \right|_{a_f = R_f(a_{-f}, s)} \stackrel{(\text{resp. } \geq)}{\geq} 0 \implies \nabla_{ds} R_f(a_{-f}, s) \stackrel{(\text{resp. } \leq)}{\leq} 0.$$

Here is the intuition behind this definition. Starting from an f -unanimous ownership matrix, consider a stock trade between two shareholders i and j of firm f in which i buys shares of another firm $g \neq f$ from j . Before the stock trade, provided firm f 's strategic plan is efficient, it maximizes the portfolio profit of each of its shareholders. The stock trade causes misalignment among firm f shareholders' interests: Shareholder i wants firm f to adjust its strategy in the direction benefiting firm g , while shareholder j wants firm f to adjust its strategy in the opposite direction.⁹ For example, under Bertrand competition with differentiated products, i will want firm f to price less aggressively while j will want it to price more aggressively. Firm f 's strategic plan is rank-preserving if, in response to the stock trade, firm f adjusts its strategy in the direction preferred by the larger shareholder involved in the trade.

Proposition 3 characterizes rank-preserving strategic plans.

Proposition 3. Consider a firm $f \in M$.

- (i) Assume that R_f is WAPP. Then, R_f is rank-preserving if and only if for every s_{*f} and every pair of firm f shareholders $i, j \in N_f(s_{*f})$, $s_{if} \geq s_{jf} \implies \gamma_{if}(s_{*f}) \geq \gamma_{jf}(s_{*f})$.

⁸Alternatively, the analysis goes through if (i) we allow the number of investors n to vary and require that $R_f(a_{-f}, s) = R_f(a_{-f}, s')$ for every a_{-f}, s, s' such that $N_f(s_{*f}) = N_f(s'_{*f})$ and $s_{i*} = s'_{i*}$ for every shareholder $i \in N_f(s_{*f})$ and (ii) interpret $\psi \neq (|\widetilde{N}| + 1)^{-1}$ as follows: Shareholders in $N_g(s_{*g}) \setminus N_f(s_{*f})$, who exert no control over firm f , participate in the stock trade to ensure ds points inside S . Notice that given any s_{*f} , we can construct $(s_{*g})_{g \neq f}$ such that s is f -unanimous and $N_g(s_{*g}) \setminus N_f(s_{*f}) \neq \emptyset$. However, this would complicate notation.

⁹More generally, when R_f is inefficient, firm f may not maximize the portfolio profit of each of its shareholders even when their interests are perfectly aligned (i.e., s is f -unanimous). Still, the stock trade causes shareholder i (resp. j) to be more (resp. less) willing to accept an adjustment in firm f 's strategy in the direction benefiting firm g .

- (ii) Assume that R_f is NB. Then, R_f is rank-preserving if and only if for every s_{*f} and every pair of firm f shareholders $i, j \in N_f(s_{*f})$, $s_{if} \geq s_{jf} \implies \beta_{if}(s_{*f})/s_{if} \geq \beta_{jf}(s_{*f})/s_{jf}$.

If firm f 's strategic plan is WAPP, it is rank-preserving if and only if the control power function γ_{*f} (weakly) preserves the ranking of shareholders in terms of the number of firm f shares they hold. One may instinctively think that under NB, $s_{if} \geq s_{jf}$ implying $\beta_{if}(s_{*f}) \geq \beta_{jf}(s_{*f})$ would be sufficient to capture the idea that larger shareholders exercise more control. However, this is not the case. For example, if $\beta_{if}(s_{*f}) = s_{if}$ for every shareholder i , then every shareholder has the same control power over firm f in terms of how firm f 's strategy changes in response to stock trades around an f -unanimous ownership matrix. Under NB, the role of γ_{if} is assumed by β_{if}/s_{if} , *not* β_{if} . The division by s_{if} captures the fact that larger shareholders have more to lose in case of disagreement. Therefore, larger shareholders have more control over firm f than smaller ones if and only if their β 's more than compensate for the fact that they have more to lose in case of disagreement.

3.3.2 Stock-trade monotonicity

Before defining the second monotonicity property, *stock-trade monotonicity*, we need to define f -neutral stock trades. An f -neutral stock trade does not make firm f want to change its strategy.

Definition 8. Fix an f -unanimous ownership matrix s . Starting from s , a $(\psi, g, i, \widetilde{N})$ -stock trade is f -neutral if for any strategy profile a_{-f} of the other firms, firm f 's conduct does not change in response to the stock trade, that is, $\nabla_{ds} R_f(a_{-f}, s) = 0$.

In an f -neutral $(\psi, g, i, \widetilde{N})$ -stock trade, ψ captures the control power of shareholder i over firm f relative to the collective control power over firm f of group $\widetilde{N}$ of shareholders. When ψ is high, the stock trade leads to a large decrease in the stakes of group $\widetilde{N}$ of shareholders in firm g and only a small increase in shareholder i 's stake in firm g . Consequently, shareholder i has a *mild* desire for firm f to adjust its strategy in the direction benefiting firm g , while group $\widetilde{N}$ of shareholders have a *strong* desire for firm f to adjust its strategy in the opposite direction. The fact that firm f 's conduct remains unchanged reveals that shareholder i exerts a lot of control over firm f relative to the control exerted by group $\widetilde{N}$ of shareholders. Intuitively, $\psi = 1$ means that group $\widetilde{N}$ of shareholders exerts no control over firm f . $\psi = 0$ means that shareholder i exerts no control over firm f . When neither i nor group $\widetilde{N}$ exert any control over firm f , a $(\psi, g, i, \widetilde{N})$ -stock trade is f -neutral for any $\psi \in [0, 1]$.

Lemma 1 characterizes two types of f -neutral stock trades: (i) those where a firm f shareholder buys firm g shares and every other firm f shareholder sells firm g shares and

(ii) those where a firm f shareholder buys firm g shares and another firm f shareholder sells firm g shares.

Lemma 1. Starting from an f -unanimous ownership matrix s ,

(i) if R_f is WAPP,

(a) a $(\psi, g, i, N_f(s_{*f}) \setminus \{i\})$ -stock trade is f -neutral if and only if $\gamma_{if}(s_{*f}) = \psi$, and

(b) a (ψ, g, i, j) -stock trade is f -neutral if and only if $\gamma_{if}(s_{*f})(1 - \psi) = \gamma_{jf}(s_{*f})\psi$,

(ii) if R_f is NB,¹⁰

(a) a $(\psi, g, i, N_f(s_{*f}) \setminus \{i\})$ -stock trade is f -neutral if and only if

$$\frac{\beta_{if}(s_{*f})/s_{if}}{\sum_{j \in N_f(s_{*f})} \beta_{jf}(s_{*f})/s_{jf}} = \psi, \text{ and}$$

(b) a (ψ, g, i, j) -stock trade is f -neutral if and only if

$$\frac{\beta_{if}(s_{*f})}{s_{if}}(1 - \psi) = \frac{\beta_{jf}(s_{*f})}{s_{jf}}\psi.$$

Lemma 1 characterizes what control weights capture. Under WAPP, γ_{if} captures shareholder i 's control power over firm f in the following way: If shareholder i 's stake in firm g increases at rate $(1 - \gamma_{if})$, while the stake in firm g of every other firm f shareholder decreases at rate γ_{if} , firm f 's conduct does not change. In this scenario, when γ_{if} is high, the other shareholders decrease their interests in firm g by a lot, which should push firm f to adjust its strategy opposite the direction that would enhance firm g 's profits. However, a small increase in shareholder i 's interests in firm g counteracts this effect, leaving firm f 's conduct unchanged. This means that shareholder i exercises a lot of control over firm f . Given Proposition 3, one can expect that in Lemma 1, the role of γ_{if} under WAPP is assumed by β_{if}/s_{if} under NB. Indeed, this is the case with one additional caveat: β_{if}/s_{if} has to be normalized. Under NB, $\beta_{if}/s_{if}/(\sum_{j \in N_f(s_{*f})} \beta_{jf}/s_{jf})$ captures shareholder i 's control power over firm f . Similarly, under WAPP, the ratio γ_{if}/γ_{jf} captures shareholder i 's control power over firm f relative to shareholder j 's control power over it. Under NB, the corresponding ratio is $\beta_{if}/s_{if}/(\beta_{jf}/s_{jf})$.

Having defined f -neutral stock trades, we are ready to define stock-trade monotone strategic plans.

Definition 9. Firm f 's strategic plan has stock-trade monotone control if for any firm $g \neq f$, any pair of shareholders $i, j \in N$, any f -unanimous ownership matrix s , any $t \in [0, \min_{g \in M: s_{ig} > 0} \min\{(1 - s_{ig})/s_{ig}, s_{jg}/s_{ig}\}]$,¹¹ and any $\psi, \psi' \in [0, 1]$, if

¹⁰In the statements below, for any $k \notin N_f(s_{*f})$, read $\beta_{kf}(s_{*f})/s_{kf} = 0$.

¹¹The constraint on t guarantees that $s'_{i*} \leq (1, 1, \dots, 1)$ and $s'_{j*} \geq (0, 0, \dots, 0)$.

(i) starting from s , a $(\psi, g, i, N_f(s_{*f}) \setminus \{i\})$ -stock trade is f -neutral, and

(ii) starting from s' , a $(\psi', g, i, N_f(s'_{*f}) \setminus \{i\})$ -stock trade is f -neutral,

then $\psi' \geq \psi$, where $s'_{k*} := s_{k*}$ for every $k \neq i, j$, $s'_{i*} := (1+t)s_{i*}$, and $s'_{j*} := s_{j*} - ts_{i*}$.

Starting from an f -unanimous ownership matrix, consider a stock trade where shareholder i buys shares from shareholder j , thereby growing her stake in *every* firm by $t \times 100\%$. Firm f 's strategic plan is stock-trade monotone if, in response, shareholder i 's control power over firm f —as measured through an f -neutral stock trade between her and every other shareholder of firm f —increases. Stock-trade monotonicity is consistent with the idea that the more firm f shares shareholder i has, the more control she exerts over firm f . It is also consistent with the idea that the more shares i has of *other* firms in the industry, the closer attention she will pay to the industry and, thus, the more influence she will have over firm f 's strategy, as discussed in section 4.1.

Proposition 4 characterizes stock-trade monotone strategic plans.

Proposition 4. Consider a firm $f \in M$.

- (i) Assume that R_f is WAPP. Then, R_f has stock-trade monotone control if and only if for every s , every pair of firm f shareholders $i, j \in N_f(s_{*f})$, and every $t \in [0, \min\{s_{jf}, 1 - s_{if}\}]$, $\gamma_{if}(s_{*f} + t(\mathbf{e}_i - \mathbf{e}_j)) \geq \gamma_{if}(s_{*f})$.
- (ii) Assume that R_f is NB. Then, R_f has stock-trade monotone control if and only if for every s , every pair of firm f shareholders $i, j \in N_f(s_{*f})$, and every $t \in [0, \min\{s_{jf}, 1 - s_{if}\}]$,

$$\frac{\beta_{if}(s'_{*f})/s'_{if}}{\sum_{k \in N_f(s_{*f})} \beta_{kf}(s'_{*f})/s'_{kf}} \geq \frac{\beta_{if}(s_{*f})/s_{if}}{\sum_{k \in N_f(s_{*f})} \beta_{kf}(s_{*f})/s_{kf}},$$

where $s'_{*f} := s_{*f} + t(\mathbf{e}_i - \mathbf{e}_j)$.

As anticipated given Lemma 1, under WAPP, shareholder i 's control over firm f increases when she buys firm f shares if γ_{if} increases. Under NB, the role of γ_{if} is assumed by $\beta_{if}/s_{if}/\sum_{k \in N_f(s_{*f})} \beta_{kf}/s_{kf}$.

3.4 Independence of irrelevant shareholders

In this section, I assume $n \geq 4$ and constrain s to lie in $\{s \in S : |N_f(s_{*f})| \geq 3\}$. I characterize the generalization of WAPP with proportional control positing that there exists δ such that $\gamma_{if}(s_{*f}) = \delta(s_{if})/\sum_{j \in N} \delta(s_{jf})$. Some potentially restrictive implications of this generalization of WAPP with proportional control—and thus of WAPP with proportional control itself—are easy to see but do not meaningfully assist us in evaluating the plausibility of the formulation. For example, assuming $\gamma_{if}(s_{*f}) = \delta(s_{if})/\sum_{j \in N} \delta(s_{jf})$

does not allow for the firm's strategic plan to maximize the portfolio profit of the absolute majority shareholder when such shareholder exists. However, many large firms with market power have no absolute majority shareholder. A characterization of this formulation of WAPP is thus useful for fully evaluating the restrictions it imposes. Proposition 5 shows that WAPP admits such a representation if and only if it satisfies three conditions.

The first condition is anonymity, which requires that the identity of shareholders not matter for firm strategy. Namely, permuting the ownership matrix s does not change the strategies prescribed by the firm's strategic plan. For example, if all of shareholder i 's shares across all firms are transferred to shareholder j , and all of shareholder j 's shares are transferred to shareholder i , the firm will choose the same strategy as it did before the transfer.

Definition 10. A permutation matrix is an $n \times n$ matrix where (i) each row has exactly one entry of 1, (ii) each column has exactly one entry of 1, and (iii) all other entries 0.

Definition 11. Firm f 's strategic plan is anonymous if for any s , a_{-f} , and permutation matrix P , $R_f(a_{-f}, s) = R_f(a_{-f}, Ps)$.

The second condition is inclusivity, which requires that every shareholder of a firm exert some control over the firm. Remember that in an f -neutral (ψ, g, i, j) -stock trade, ψ captures the control power of shareholder i over firm f relative to shareholder j 's control over firm f . $\psi = 0$ (resp. $\psi = 1$) means that i (resp. j) has no control over firm f .

Definition 12. Firm f 's strategic plan is inclusive if for any firm $g \neq f$, any f -unanimous ownership matrix s , and any pair of firm f shareholders $i, j \in N_f(s_{*f})$, a $(0, g, i, j)$ -stock trade is not f -neutral.

Remark: After relabeling of the shareholders, this also implies that a $(1, g, i, j)$ -stock trade is not f -neutral either.

The third condition is independence of irrelevant shareholders, which requires that the relative control power of two shareholders over firm f be unaffected by stock trades between other shareholders of the firm.

Definition 13. Firm f 's strategic plan satisfies independence of irrelevant shareholders (IIS) if for any $\psi \in [0, 1]$, any firm $g \neq f$, any f -unanimous ownership matrices s and s' , and any pair of shareholders $i, j \in N_f(s_{*f})$ such that $s'_{if} = s_{if}$ and $s'_{jf} = s_{jf}$, if a (ψ, g, i, j) -stock trade is f -neutral starting from s , then it is f -neutral also starting from s' .

Remark: Given any pair $i, j \in N$ of shareholders, IIS does not impose any conditions on the relation between (i) the relative control power over firm f of the two shareholders under s_{*f} such that $i, j \in N_f(s_{*f})$ and $|N_f(s_{*f})| = 2$ and (ii) the control power over firm f of the two shareholders under s'_{*f} such that $i, j \in N_f(s'_{*f})$ and $|N_f(s'_{*f})| \geq 3$. This

is because if $|N_f(s_{*f})| = 2$ and $|N_f(s'_{*f})| \geq 3$, $s'_{if} \neq s_{if}$ or $s'_{jf} \neq s_{jf}$. To simplify the exposition, we have constrained s to lie in $\{s \in S : |N_f(s_{*f})| \geq 3\}$. The assumption $n \geq 4$ plays a similar role: if $n = 3$, then $|N_f(s_{*f})| = 3$, so fixing s_{if} and s_{jf} pins down the remaining entry of s_{*f} , and IIS relates no two distinct ownership structures of firm f .¹²

Proposition 5. Assume that R_f is WAPP. Then, R_f satisfies anonymity, inclusivity, and IIS if and only if there exists $\delta : [0,1] \rightarrow \mathbb{R}_+$ with $\delta(0) = 0$ and $\delta(x) > 0$ for every $x > 0$ such that for every s_{*f} and every $i \in N$, $\gamma_{if}(s_{*f}) = \delta(s_{if}) / \sum_{j \in N} \delta(s_{jf})$.

Remark: If R_f is also rank-preserving, δ is non-decreasing on $[0,1/2]$, since by part (i) of Proposition 3, $\delta(x) \geq \delta(y)$ whenever $x \geq y > 0$ and $x + y \leq 1$.

4 Discussion

This section discusses the (i) irrelevance of external factors, (ii) efficiency, (iii) monotonicity, (iv) anonymity, inclusivity, and independence of irrelevant shareholders conditions, as well as (v) the effects of ownership dispersion on firm conduct, (vi) potential formulations of the disagreement payoff function in NB, and (vii) the assumptions imposed in some of the analysis of section 3.

4.1 Irrelevance of external factors

What characterizes WAPP as a special case of NB is that under WAPP, the distribution of power among firm f shareholders *cannot* depend on external factors such as (i) the stakes of firm f shareholders in competing firms, (ii) the strategies of other firms, and (iii) market conditions (e.g., demand or production technology). Here, I discuss ways in which each of the three factors might affect the distribution of power among firm f shareholders. First, firm f shareholders with larger stakes in competitors may pay closer attention to the industry and thus exert more control over firm f . There is theoretical and empirical support that investors pay closer attention to a stock when that stock is a larger part of their portfolios (Van Nieuwerburgh and Veldkamp, 2010; Fich et al., 2015; Iliev et al., 2021), and this can mediate the impact of common ownership on firm conduct (Gilje et al., 2020). Second, consider shareholder i of firm f who also has shares of competing firm g . Shareholder i will have stronger incentives to exert control over firm f when firm g 's strategy is such that firm f has a lot of room to influence g 's profitability. That is, because

¹²These assumptions hardly limit the scope of the analysis given that in practice, if a firm f has some shareholders who also have stakes in competitors, then firm f rarely has exactly two shareholders. Also, if there are only two shareholders and one owns more than half the shares, it is reasonable to assume that the firm will pursue the majority shareholder's interests. If we allow for any $s \in S$, in Proposition 5, we would need a separate function $\delta_2 : [0,1] \rightarrow \mathbb{R}_+$ such that for every s_{*f} with $|N_f(s_{*f})| = 2$ and every $i \in N$, $\gamma_{if}(s_{*f}) = \delta_2(s_{if}) / \sum_{j \in N} \delta_2(s_{jf})$. Both δ_2 and δ would be able to deal with s_{*f} with $|N_f(s_{*f})| = 1$ with $\delta_2(1) \neq 0$ or $\delta(1) \neq 0$ arbitrarily defined.

firm g 's strategy can affect how consequential firm f 's conduct is for shareholder i 's wealth. For instance, if firm g expands its production capacity and orders large input quantities to scale up production, firm f 's effect through its pricing and production strategy on firm g 's profit margin will have a large impact on firm g 's profits. In that case, shareholder i will have strong incentives to push firm f to not be very aggressive. On the other hand, if firm g scales back production or even exits a market that firm f operates in, firm f will have limited room to affect firm g 's profitability, so shareholder i 's incentives will be less strong. Third, firm f shareholders who also have shares of competing firm g may have stronger incentives to exert control over firm f when firm f and g 's goods are strong substitutes, since in that case firm f 's strategy will have a more pronounced impact on firm g 's profits.

Majority voting is NB but not WAPP. WAPP is a natural benchmark, since aggregating preferences using Pareto weights is standard in the economics literature. Also, it seems natural that these weights will depend on the firm's ownership structure. It is less natural within the GWAPP framework to directly specify how the weights may depend on external factors. However, requiring the weights to be independent of external factors rules out other natural benchmarks. For example, let A_f be an open real interval, fix some π , constrain attention to pure strategies, and assume $u_i(a_f, a_{-f}, s_{i*})$ is differentiable and single-peaked in a_f . Let $R_f(a_{-f}, s)$ be the median shareholder's most preferred strategy, arising from majority voting (e.g., as in Chiappinelli et al. (2023)).¹³ This R_f is strongly efficient and internally consistent, since it satisfies the conditions of Definition 3 for $\widetilde{N}(s_{*f}) = N_f(s_{*f})$. Therefore, by Proposition 1, it is NB. However, it is not WAPP.

Here is why. Assume by contradiction that R_f is WAPP, and fix some a_{-f} . First, observe that for every s and i , $\gamma_{if}(s_{*f}) = 0$ if i is *not* a median shareholder. To see this, take any s such that not all firm f shareholders have the same most preferred strategy and any $i \in N_f(s_{*f})$ that is not a median shareholder. There exists s' with $s_{j*} = s'_{j*}$ for every $j \in N_f(s_{*f}) \setminus \{i\}$, $s_{if} = s'_{if}$, and $s_{i*} \neq s'_{i*}$, such that the preferences of the median firm f shareholder under s are the same as under s' and thus $R_f(a_{-f}, s) = R_f(a_{-f}, s')$, but i 's preferences under s_{i*} are different from her preferences under s'_{i*} .¹⁴ $R_f(a_{-f}, s) = R_f(a_{-f}, s')$

¹³More than 90% of S&P 500 companies follow a majority voting standard for uncontested director elections (see <https://www.conference-board.org/publications/corporate-board-practices-2021-edition>).

¹⁴For this and following arguments, some minimal conditions on π evaluated at a_{-f} are required. For example, under minimal conditions on π , i 's preferences under s_{i*} are indeed different from her preferences under s'_{i*} for some $s'_{i*} \neq s_{i*}$ such that the preferences of the median firm f shareholder under s are the same as under s' (e.g., for some a_{-f} and rival g , a shareholder i with relative stake $\lambda_{i,fg}$ in g has a preferred strategy for firm f that differs from the preferred strategy of every shareholder j with $\lambda_{j,fg} > \lambda_{i,fg}$).

implies

$$\arg \max_{a_f \in A_f} \left\{ \sum_{j \in N_f(s_{*f})} \gamma_{jf}(s_{*f}) u_j(a_f, a_{-f}, s_{j*}) \right\} = \arg \max_{a_f \in A_f} \left\{ \sum_{j \in N_f(s'_{*f})} \gamma_{jf}(s'_{*f}) u_j(a_f, a_{-f}, s'_{j*}) \right\}$$

or equivalently, given that $s_{*f} = s'_{*f}$ and $s_{j*} = s'_{j*}$ for every $j \in N_f(s_{*f}) \setminus \{i\}$,

$$\begin{aligned} & \arg \max_{a_f \in A_f} \left\{ \sum_{j \in N_f(s_{*f})} \gamma_{jf}(s_{*f}) u_j(a_f, a_{-f}, s_{j*}) \right\} \\ &= \arg \max_{a_f \in A_f} \left\{ \gamma_{if}(s_{*f}) u_i(a_f, a_{-f}, s'_{i*}) + \sum_{j \in N_f(s_{*f}) \setminus \{i\}} \gamma_{jf}(s_{*f}) u_j(a_f, a_{-f}, s_{j*}) \right\}, \end{aligned}$$

which gives $\gamma_{if}(s_{*f}) = 0$.

Now, notice that there exist s and s' such that $s_{*f} = s'_{*f}$ and $M_f(s) \cap M_f(s') = \emptyset$, where $M_f(s)$ is the set of median shareholders under s and $M_f(s')$ is the set of median shareholders under s' . For example, there can be a unique median shareholder under s and a different unique median shareholder under s' while $s_{*f} = s'_{*f}$. We have then that for every $i \notin M_f(s)$, $\gamma_{if}(s_{*f}) = 0$, and for every $i \notin M_f(s')$, $\gamma_{if}(s'_{*f}) = 0$, which given that $s_{*f} = s'_{*f}$, implies $\gamma_{if}(s_{*f}) = 0$ for every $i \in N_f(s_{*f})$, a contradiction.

We conclude that the strategic plan resulting from majority voting induces a distribution of power among shareholders that depends on external factors. Indeed, if in the argument above, γ_{*f} was allowed to depend on the whole ownership matrix s , no contradiction would arise. Similar arguments can be made when instead of varying s , we vary a_{-f} or π , while we could also allow for a voting cost.

4.2 Efficiency

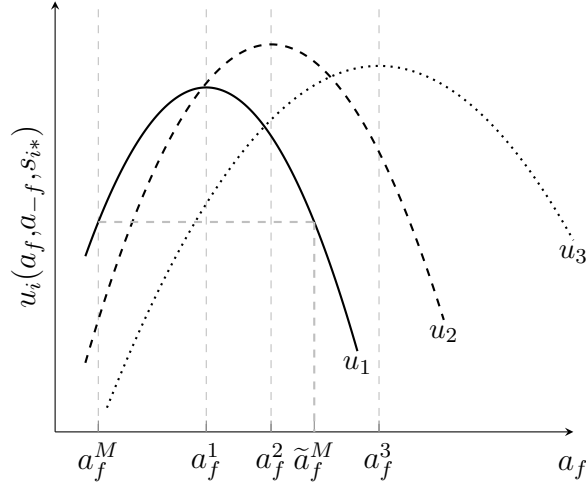
In this paper, I have studied flexible representations of firm conduct that can be embedded in theoretical and empirical studies of oligopolistic competition under common ownership. In doing so, I focused more on the problem of aggregating shareholders' preferences than on whether and through what mechanisms the firm's management is induced to pursue the shareholders' aggregated preferences despite agency frictions. Indeed, a careful examination of specific channels through which common ownership can affect firm conduct despite agency frictions is a promising avenue for research (see, e.g., Hemphill and Kahan, 2020; Antón et al., 2023). Still, the representations of firm conduct I derive can apply in the presence of agency frictions. For example, as Backus et al. (2020) observe, the measure of common ownership proposed by Gilje et al. (2020) that accounts for investor attention corresponds to a specific formulation of γ_{*f} in WAPP.

However, if strong enough, agency frictions could make the firm's strategic plan inefficient, and thus not NB or WAPP. Managers may engage in "empire-building." To

enhance their power and reputation, entrenched managers may over-invest and over-produce (compared to the profit-maximizing levels) and make value-destroying acquisitions, expanding the firm beyond its optimal size (Jensen, 1986; Masulis et al., 2007; Harford et al., 2012). Therefore, the management’s preferred strategy may be inefficient; namely, it can be too aggressive (e.g., in pricing) even for the firm’s shareholders who have no stakes in competing firms. In that case, inefficiencies will arise if the management brings firm strategy close enough to its preferred strategy. This is shown in Figure 1, where firm f has three shareholders, a_f^i is shareholder i ’s preferred strategy, and a_f^M is the management’s preferred strategy. If $R_f(a_{-f}, s)$ is close to a_f^M so that $R_f(a_{-f}, s) < a_f^1$, R_f is inefficient.

WAPP and NB can account for inefficiencies of this kind if the firm’s management is treated as a shareholder. For example, Azar and Ribeiro (2022) employ a modification of WAPP that allows for managerial entrenchment. They assume that the manager prefers to maximize the firm’s own profit, bringing the firm’s objective closer to own profit maximization. It can be shown that Azar and Ribeiro’s (2022) model is equivalent to assigning control power γ_f^m and “cash-flow right” s_f^m normalized to $s_f^m = 1$ to the manager of firm f (so that $s_f^m + \sum_{i \in N} s_{if} = 2$). Ownership diffusion has similar effects in their model as in standard WAPP (see section 4.5): As ownership becomes dispersed, the manager has more power and thus internalizes the shareholders’ interests to a lesser degree. When all shareholders are atomistic, the manager maximizes the firm’s own profit—even if all shareholders are completely diversified across the industry.

Figure 1: Inefficient strategic plan due to management entrenchment



Note: a_f^i is shareholder i ’s preferred strategy. a_f^M is firm f ’s management’s preferred strategy. $\tilde{a}_f^M \neq a_f^M$ is such that $u_1(\tilde{a}_f^M, a_{-f}, s_{1*}) = u_1(a_f^M, a_{-f}, s_{1*})$.

4.3 Monotonicity

The monotonicity properties proposed here are not too strong. By studying firm f ’s strategic plan around f -unanimous matrices, we allow firm f shareholders to be het-

erogeneous in only two dimensions: (i) their size and (ii) their identity. Thus, we can expect rank preservation to be satisfied unless shareholder identity plays a strong enough role to overturn the shareholder size effect, resulting in a smaller shareholder exerting more control than a larger one. Particularly, if firm f 's strategic plan is anonymous, we can expect rank preservation to be satisfied. At an ownership matrix s that is not f -unanimous, firm f shareholders are heterogeneous also (iii) in terms of their preferences over firm f strategies (since $\lambda_{i,f*}$ varies across $i \in N_f(s_{*f})$). This heterogeneity, in turn, implies that (iv) competing firms' strategies and (v) market conditions can have heterogeneous effects on different shareholders' control over firm f . For a discussion of how shareholders' stakes in competing firms can interact with competitors' strategies and market conditions to affect the distribution of power among shareholders, see section 4.1. A stronger rank preservation condition defined around not necessarily f -unanimous matrices would be violated if the combined effect of external factors and shareholder identity on the distribution of power among shareholders is strong enough to dominate the shareholder size effect. Stock-trade monotonicity suggests that firm f shareholder i 's control over firm f (weakly) increases if i increases her stake in firm f while *no other* shareholder increases her stake in firm f . Given that it imposes no conditions on the relative control power of shareholders, stock-trade monotonicity is unlikely to be violated due to potential effects of shareholder identity.

4.4 Anonymity, inclusivity, and independence of irrelevant shareholders

The generalization of WAPP with proportional control positing that there exists δ such that $\gamma_{if}(s_{*f}) = \delta(s_{if}) / \sum_{j \in N} \delta(s_{jf})$ imposes three conditions (in addition to efficiency and irrelevance of external factors imposed by WAPP) on the firm's strategic plan: anonymity, inclusivity, and independence of irrelevant shareholders. Here I discuss how each of the three conditions might be violated. Anonymity may fail if different investors have different expertise or power to influence firm strategy. For example, a venture capitalist may have more resources than a retail investor or stronger incentives than an index fund (with the same number of firm f shares) to exercise control over firm f , and thus have a stronger influence on firm strategy. To be sure, while some argue that index funds exert limited control (Bebchuk and Hirst, 2019; Heath et al., 2022), there is evidence suggesting otherwise (Appel et al., 2016; Fisch et al., 2019; Shekita, 2022; Lewellen and Lewellen, 2022). Inclusivity may, for example, fail if there are significant fixed costs in exerting control that prevent smaller shareholders from doing so. Indeed, there is evidence pointing in this direction (Gantchev, 2013; Iliev and Lowry, 2015; Brav et al., 2022). Last, IIS may fail if shareholder i 's control power relative to shareholder j 's control power over firm f depends on the pivotality of each of the two shareholders in voting, which in turn depends on the number of firm f shares held by all other shareholders. Zingales (1994, 1995) finds

evidence pointing in this direction—namely, that the value of voting rights increases with pivotality.

4.5 Powerlessness of diffuse ownership

In this section, I discuss the effects of ownership dispersion under WAPP and NB. Before doing so, I define an f -bianimous ownership matrix. An ownership matrix is f -bianimous if firm f shareholders can be divided into two groups such that within each group, all shareholders have aligned interests.

Definition 14. An ownership matrix s is called f -bianimous if there exists a partition $\{N_1, N_2\}$ of $N_f(s_{*f})$ such that for every $i, j \in N_f(s_{*f})$, if $(i, j) \in N_1^2$ or $(i, j) \in N_2^2$, then $\lambda_{i;f*} \equiv s_{i*}/s_{if} = s_{j*}/s_{jf} \equiv \lambda_{j;f*}$.

We will study how the firm adjusts its strategy as ownership within group N_2 of its shareholders becomes dispersed.

Ownership diffusion under WAPP. Fix some $i_1 \in N_1$ and $i_2 \in N_2$. For any firm $g \neq f$,

$$\lambda_{fg}(s) = \frac{\lambda_{i_1;fg} \sum_{i \in N_1} \gamma_{if}(s_{*f}) s_{if} + \lambda_{i_2;fg} \sum_{i \in N_2} \gamma_{if}(s_{*f}) s_{if}}{\sum_{i \in N_1} \gamma_{if}(s_{*f}) s_{if} + \sum_{i \in N_2} \gamma_{if}(s_{*f}) s_{if}}.$$

Consider a sequence $s(\nu)_{\nu \in \mathbb{N}}$ of f -bianimous ownership matrices such that $\lambda_{i_1;fg}$, $\lambda_{i_2;fg}$, N_1 , and s_{if} are fixed along the sequence for every $i \in N_1$, but the holdings of shareholders in N_2 are divided across more and more shareholders, so that $|N_2| \rightarrow \infty$ and $\max_{i \in N_2} s_{if} \rightarrow 0$. Then, $\sum_{i \in N_2} \gamma_{if}(s_{*f}) s_{if} \rightarrow 0$, so $\lambda_{fg}(s) \rightarrow \lambda_{i_1;fg}$ (i.e., the weight firm f assigns on firm g 's profit converges to the weight group N_1 of shareholders assigns to it) unless $\sum_{i \in N_1} \gamma_{if}(s_{*f}) s_{if} \rightarrow 0$ at the same or faster rate than $\sum_{i \in N_2} \gamma_{if}(s_{*f}) s_{if} \rightarrow 0$. If $\lambda_{fg}(s) \rightarrow \lambda_{i_1;fg}$ for every firm $g \neq f$, then, under standard assumptions, firm f 's strategy will converge to the strategy most preferred by the group N_1 of shareholders.¹⁵ For $\sum_{i \in N_1} \gamma_{if}(s_{*f}) s_{if} \rightarrow 0$, it is necessary that $\max_{i \in N_1} \gamma_{if}(s_{*f}) \rightarrow 0$. In that case, the shareholders' ranking can be preserved only weakly in the limit (see Proposition 3). This means that under WAPP, at least one of the following must hold: (i) diffuse ownership is powerless in the sense that atomistic shareholders exert no control over the firm (given that in the limit, firm f only pursues group N_1 's interests) or (ii) when some of firm f shareholders are atomistic, the control power of every individual shareholder—whether she is large or atomistic—vanishes. This suggests that under WAPP, there is a tension

¹⁵For example, if A_f is compact and π_g is continuous for every firm g , then by Berge's Maximum Theorem, $R_f(a_{-f}, s) \equiv \arg \max_{a_f} \{\pi_f(a_f, a_{-f}) + \sum_{g \in M \setminus \{f\}} \lambda_{fg}(s) \pi_g(a_f, a_{-f})\}$ is upper hemicontinuous in λ_{f*} , so the limit of $R_f(a_{-f}, s)$ as $\lambda_{f*}(s) \rightarrow \lambda_{i_1;f*}$ (given that it exists) is a subset of $\arg \max_{a_f} \{\pi_f(a_f, a_{-f}) + \sum_{g \in M \setminus \{f\}} \lambda_{i_1;fg} \pi_g(a_f, a_{-f})\}$.

between (i) allowing for atomistic shareholders to collectively exert control over the firm and (ii) allowing for large shareholders to have control power.

With additional structure imposed on WAPP, the tension becomes more stark. Let $\gamma_{if}(s_{*f}) = \delta(s_{if}) / \sum_{j \in N} \delta(s_{jf})$ with

$$\delta(s_{if}) = \begin{cases} s_{if}^\zeta & \text{if } s_{if} > 0 \\ 0 & \text{if } s_{if} = 0 \end{cases}$$

for $\zeta \geq 0$, as in Backus et al. (2021a), Antón et al. (2023), and Ederer and Pellegrino (2025). If $\zeta > 0$, $\lambda_{fg}(s) \rightarrow \lambda_{i1;fg}$ as $|N_2| \rightarrow \infty$ and $\max_{i \in N_2} s_{if} \rightarrow 0$. If $\zeta = 0$, firm f maximizes the *unweighted* average of its shareholders' portfolio profits, and thus, it assigns weight

$$\lambda_{fg}(s) = \frac{\sum_{i \in N_f(s_{*f})} \gamma_{if}(s_{*f}) s_{ig}}{\sum_{i \in N_f(s_{*f})} \gamma_{if}(s_{*f}) s_{if}} = \frac{\sum_{i \in N_f(s_{*f})} s_{ig} / |N_f(s_{*f})|}{\sum_{i \in N_f(s_{*f})} s_{if} / |N_f(s_{*f})|} = \sum_{i \in N_f(s_{*f})} s_{ig}$$

to firm g 's profit. Therefore, if $\zeta = 0$, atomistic shareholders collectively exert control over firm f , since λ_{fg} represents all shareholders' interests and remains fixed as ownership by N_2 is diffused. However, at the same time, λ_{fg} is unreasonably high. It is equal to 1 if firm g 's shareholders are a subset of firm f shareholders. To see why this is unrealistic, start from $s = I_n$, where I_n is the identity matrix (*i.e.*, each firm is owned by a unique shareholder).¹⁶ If we slightly perturb s , so that each shareholder has some shares of every firm, the firms will switch from own profit maximization to each maximizing aggregate industry profits—making the multiplant monopolist's solution an equilibrium—a stark discontinuity. The *unweighted* average of the firm's shareholders' portfolio profits can arise as the firm's objective absent agency and trading frictions. Absent such frictions, firm strategy could maximize the shareholders' aggregate portfolio profits, who can afterwards potentially make monetary transfers among themselves.

Ownership diffusion under NB. Fix some s , a_{-f} , $i_1 \in N_1$, and $i_2 \in N_2$, and assume that for every a_{-f} , s , and $i \in N$, $d_{if}(a_{-f}, s) = u_i(\alpha_f^d(a_{-f}, s), a_{-f}, s_{i*})$, where $\alpha_f^d(a_{-f}, s)$ the strategy chosen by firm f in case of disagreement. Then, firm f 's objective is

$$\begin{aligned} & \prod_{i \in N_f(\beta_{*f}(s_{*f}))} (u_i(a_f, a_{-f}, s_{i*}) - d_{if}(a_{-f}, s))^{\beta_{if}(s_{*f})} \\ & \propto (u_{i_1}(a_f, a_{-f}, s_{i_1*}) - d_{i_1f}(a_{-f}, s))^{\sum_{i \in N_1} \beta_{if}(s_{*f})} \\ & \quad \times (u_{i_2}(a_f, a_{-f}, s_{i_2*}) - d_{i_2f}(a_{-f}, s))^{\sum_{i \in N_2} \beta_{if}(s_{*f})}. \end{aligned}$$

¹⁶The same argument can be made if we start with each firm owned by multiple shareholders who do not own shares of other firms.

Therefore, if we consider a sequence $s(\nu)_{\nu \in \mathbb{N}}$ of f -bianimous ownership matrices such that $|N_2| \rightarrow \infty$ and $\max_{i \in N_2} s_{if} \rightarrow 0$, the following two can hold at the same time: (i) firm f 's strategy does *not* converge to the strategy most preferred by the group N_1 of shareholders and (ii) $\sum_{i \in N_1} \beta_{if}(s_{*f}(\nu))$ is bounded away from 0. Thus, NB can relax the tension that arises under WAPP between (i) allowing for atomistic shareholders to collectively exert control over the firm and (ii) allowing for large shareholders to have control power.

4.6 Disagreement payoffs

The choice of the disagreement payoff function can be guided by empirical evidence. For example, in testing alternative conduct models as in Backus et al. (2021b), one can test which of several plausible disagreement payoff functions is most consistent with the data. Here, I discuss some potential formulations of the disagreement payoff function. A combination of those formulations is also worth considering (e.g., shareholder i 's disagreement payoff is her expected portfolio profit from a lottery between the different formulations).

Management-preferred strategy. In case of disagreement, firm f 's management may be able to implement their own most preferred strategy, unconstrained by the shareholders' interests. This can, for example, be the strategy that maximizes firm f 's profit or the one that maximizes f 's revenue.¹⁷ In the latter case, and as discussed in section 4.2, the management's preferred strategy can be inefficient; namely, it can be too aggressive even for the firm's shareholders who have no stakes in competing firms. This would create incentives for shareholders to reach an agreement. For instance, in the example of Figure 1, if a_f^M is chosen in case of disagreement and $\beta_{1f}(s_{*f}) > 0$, then $R_f(a_{-f}, s) \in [a_f^1, \tilde{a}_f^M)$.

Status-quo. In case of disagreement, the firm implements some status-quo strategy. The status-quo could, for example, entail no substantial new projects, R&D, or increases in production capacity. In a dynamic setting, it could mean that the firm repeats last period's strategy. A more extreme formulation would have the firm not produce at all in case of disagreement.

Random dictatorship. The Online Appendix describes the random dictatorship disagreement payoffs. Under this formulation, there exists a lottery weight function $\mu_{*f} : \Delta^n \rightarrow \Delta^n$ such that, in case of disagreement, for every $i \in N_f(s_{*f})$, shareholder i gets to implement her most preferred strategy with probability $\mu_{if}(s_{*f})$. While like

¹⁷If in case of disagreement the strategy that maximizes firm f 's profit is implemented, and there is some shareholder i of firm f with shares in no other firms and $\beta_{if} > 0$, then NB will prescribe the strategy that maximizes firm f 's profit, even if s_{if} is small. Therefore, one needs to be careful when specifying the disagreement payoff function.

WAPP, NB with disagreement payoffs derived from random dictatorship can be best interpreted as an as-if assumption, this formulation of disagreement payoffs has certain desirable properties. First, through the lottery weights μ_{*f} , it can account for the relative power of shareholders. Second, when A_f is convex and the portfolio profit $u_i(a_f, a_{-f}, s_{i*})$ of every shareholder $i \in N_f(s_{*f})$ is strictly concave in a_f ,¹⁸ by Jensen's inequality, there will be strict benefits from agreement—as long as there exist shareholders i and j with $\mu_{if}(s_{*f}), \mu_{jf}(s_{*f}) > 0$ who have distinct most preferred strategies. Last, disagreement payoffs derived from random dictatorship have connections to strategic foundations of Nash bargaining. Howard (1992) describes a game whose unique perfect equilibrium outcome implements symmetric NB with random dictatorship disagreement payoffs.

4.7 Assumptions used in section 3

For some of the results, for tractability, I have restricted attention to the case where A_f is a real interval and firm f 's conduct around f -unanimous matrices is characterized by the FOC. This does not limit the scope of the results. Given that the firm's strategic plan needs to specify firm conduct across a range of environments, it is informative to study the strategic plan in this simple setting. The important point is that after a $(\psi, g, i, \tilde{N})$ -stock trade, shareholder i wants firm f to adjust its strategy in the direction benefiting firm g , which is the exact opposite direction from the one in which group $\tilde{N}$ of shareholders wants firm f to adjust its strategy. The analysis can be performed when A_f is multidimensional.

The restriction to the case where firm f 's conduct around f -unanimous matrices is characterized by the FOC is needed because, as discussed in section 3.3, local changes in firm conduct around an f -unanimous matrix can affect firm f 's conduct only through their effect on shareholders' interests. Without the necessary smoothness imposed on the firm's objective function around f -unanimous matrices, if for example $R_f(\alpha_{-f}, s)$ is strongly efficient and internally consistent (and thus NB) and we impose restrictions on $R_f(\alpha_{-f}, s)$ only around f -unanimous ownership matrices, the bargaining weights $\beta_{*f}(s_{*f})$ could be defined arbitrarily as long as the controlling shareholders' weights sum up to 1. Monotonicity conditions around a larger subset of the space S of ownership matrices would impose restrictions on how the distribution of power among shareholders might depend on external factors. For example, strengthening the rank preservation condition to require that starting from *any*—not necessarily f -unanimous—ownership matrix s , firm f adjust its strategy in the direction preferred by the larger shareholder involved in a $(1/2, g, i, j)$ -stock trade would not impose any additional restrictions under WAPP. On the other hand, under NB, it would impose conditions on how external factors can affect (through the disagreement payoffs) the distribution of power among shareholders.

¹⁸Lemma 3 in the Online Appendix provides sufficient conditions for strict concavity in a homogeneous product Cournot market.

5 Implications for common ownership hypothesis testing

This section shows that misspecification of the model of firm conduct can lead to an incorrect rejection of the common ownership hypothesis. Namely, when a firm's strategic plan is NB (and not WAPP) but an econometrician considers only WAPP models, she may find own profit maximization to fit firm conduct better than WAPP, and as a result, mistakenly conclude that common ownership does not affect firm conduct.

To illustrate this, I consider a setting where the firm's strategic plan (i.e., the true data-generating process) follows the median shareholder's interests, which as shown in section 4.1, is NB but not WAPP. The econometrician tests whether common ownership affects firm f 's conduct. I study a simplified form of conduct testing in a stylized setting that allows for clean analytical results. The test is similar to the firm conduct test in Backus et al. (2021b). Firm f competes with firm g in a homogeneous product Cournot duopoly with inverse demand $P(Q) = \max\{a - bQ, 0\}$, where $a, b > 0$ and Q is the total quantity. Firm f 's true marginal cost is $c_f^* \geq 0$. The econometrician knows (i.e., has accurately estimated) the demand but does not know firm f 's marginal cost. She observes a sample $\{(q_f^t, q_g^t)\}_{t=1}^T$ of T observations of the two firms' actions with $q_f^t \neq q_g^t$ for some t, t' and $q_f^t > 0$ for every t . For simplicity, the ownership structure s is held constant in the sample.¹⁹ Firm f 's strategic plan is NB, following the median shareholder's preferences: For every observation t , $q_f^t = R_f(q_g^t, s) \equiv \max\{(a - c_f^*)/(2b) - (1 + \lambda_{fg}^m)q_g^t/2, 0\}$, where $\lambda_{fg}^m := \min\left\{x \geq 0 : \sum_{j \in N_f(s_{*f}) : \lambda_{j;fg} \leq x} s_{jf} \geq \frac{1}{2} > \sum_{j \in N_f(s_{*f}) : \lambda_{j;fg} < x} s_{jf}\right\}$ is the weight firm f 's median shareholder assigns to firm g 's profit.²⁰

To test the common ownership hypothesis, the econometrician compares which of two models of firm conduct best explains firm f 's behavior. The first model estimates firm f 's marginal cost to explain firm f 's conduct as the result of own profit maximization (OPM), leading to a sum of squared errors given by $\Sigma^{\text{OPM}} := \inf_{c_f} \sum_{t=1}^T \left(q_f^t - R_f^{\text{OPM}}(q_g^t, s; c_f)\right)^2$, where $R_f^{\text{OPM}}(q_g^t, s; c_f) := (a - c_f)/(2b) - q_g^t/2$ is firm f 's own-profit-maximizing quantity given marginal cost c_f . The second model estimates firm f 's marginal cost and control power function γ_{*f} to explain firm f 's conduct as the result of a WAPP and rank-preserving strategic plan, leading to a sum of squared errors given by $\Sigma^{\text{WAPP}} := \inf_{c_f, \gamma_{*f}} \sum_{t=1}^T \left(q_f^t - R_f^{\text{WAPP}}(q_g^t, s; c_f)\right)^2$, where $R_f^{\text{WAPP}}(q_g^t, s; c_f) := (a - c_f)/(2b) - (1 + \lambda_{fg}(s))q_g^t/2$ and, given Proposition 3, γ_{*f} should satisfy $s_{if} \geq s_{jf} \implies \gamma_{if}(s_{*f}) \geq \gamma_{jf}(s_{*f})$.²¹

¹⁹Analogous results can be derived if $(s^t)_{t=1}^T$ varies in the sample (e.g., across time). The variation in firm g 's quantity can, for example, be due to variation in its cost. The Online Appendix describes how the analysis applies to a simple noisy setting where firm f experiences unobserved idiosyncratic cost shocks.

²⁰Every firm f shareholder i 's portfolio profit $u_i(q_f^t, q_g^t, s_{i*})$ is single-peaked in q_f^t , so majority voting leads to the firm following the median shareholder's preferences.

²¹For simplicity, we have allowed the econometrician to fit any $c_f \in \mathbb{R}$. If we required $c_f \geq 0$, it would be even easier for OPM to fit the data better than WAPP. To see this, notice that basent the non-negativity constraint on c_f , when firm f places weight x on g 's profit, the marginal cost that best fits firm f 's conduct is $c_f^* + b(\lambda_{fg}^m - x)\bar{q}_g$, which is less than c_f^* for $x > \lambda_{fg}^m$. Imposing $c_f \geq 0$ therefore leaves Σ^{OPM} unchanged while raising Σ^{WAPP} whenever $\lambda > \lambda_{fg}^m$ and c_f^* is small.

Let shareholders be ordered by their firm f stakes: $s_{1f} \geq s_{2f} \geq \dots \geq s_{n_f}$. Define the number $n_f := |N_f(s_{*f})|$ of firm f shareholders and for every $k \in \{1, \dots, n\}$, $\Lambda_k := \sum_{i=1}^k s_{ig} / \sum_{i=1}^k s_{if}$. Lemma 2 characterizes the possible values of the weight $\lambda_{fg}(s)$ firm f assigns to firm g 's profit under a WAPP and rank-preserving strategic plan.

Lemma 2. Take any $x \in \mathbb{R}$. There exists WAPP and rank-preserving strategic plan $R_f^{\text{WAPP}}(q_g^t, s; c_f) \equiv (a - c_f)/(2b) - (1 + \lambda_{fg}(s))q_g^t/2$ with $\lambda_{fg}(s) = x$ if and only if $x \in [\underline{\lambda}, \bar{\lambda}]$, where $\underline{\lambda} := \min_{k \in K} \Lambda_k$, $\bar{\lambda} := \max_{k \in K} \Lambda_k$, and $K := \{k \leq n_f : s_{kf} > s_{k+1,f}\}$.²²

Proposition 6 characterizes when OPM explains firm conduct better than *any* WAPP and rank-preserving strategic plan. Specific formulations of WAPP (e.g., the one assuming proportional control) generically lead to $\lambda_{fg}(s) > \underline{\lambda}$ and will thus underperform OPM under weaker conditions than the one specified in Proposition 6.

Proposition 6. OPM explains firm f 's conduct better than *any* WAPP and rank-preserving strategic plan (i.e., $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$) if and only if $\underline{\lambda} > 2\lambda_{fg}^m$.

If larger shareholders are more diversified than smaller ones (i.e., $\lambda_{1;fg} \geq \lambda_{2;fg} \geq \dots \geq \lambda_{n_f;fg}$), then $\underline{\lambda} = \Lambda_{n_f} = \sum_{i \in N_f(s_{*f})} s_{ig} = \sum_{i \in N_f(s_{*f})} s_{if} \lambda_{i;fg}$ is a weighted average of $(\lambda_{i;fg})_{i \in N_f(s_{*f})}$. If, in addition, firm g 's shareholders are a subset of firm f 's shareholders, $\underline{\lambda} = 1$, so the condition of Proposition 6 reads $\lambda_{fg}^m < 1/2$. λ_{fg}^m is a “weighted median” of $(\lambda_{i;fg})_{i \in N_f(s_{*f})}$. Therefore, OPM outperforms WAPP if a sizable number of shares is held by shareholders who are considerably more diversified than the majority of shareholders. Indeed, in practice, large shareholders such as investment funds tend to be more diversified and collectively hold sizable—but not majority—stakes in companies (Backus et al., 2021a).

Consider, for example, an ownership structure s where two groups of shareholders own firm f : $N_f(s_{*f}) = \bar{N} \cup \underline{N}$ with $\bar{N} \cap \underline{N} = \emptyset$. Group $\bar{N}$ is comprised of large shareholders with large stakes in firm g : $s_{if} = \bar{s}_{\text{own}}$ and $s_{ig} = \bar{s}_{\text{other}}$ for every $i \in \bar{N}$. Group $\underline{N}$ is comprised of smaller shareholders with small stakes in firm g : $s_{if} = \underline{s}_{\text{own}} < \bar{s}_{\text{own}}$ and $s_{ig} = \underline{s}_{\text{other}}$ with $\underline{s}_{\text{other}}/\underline{s}_{\text{own}} < \bar{s}_{\text{other}}/\bar{s}_{\text{own}}$ for every $i \in \underline{N}$. Group $\underline{N}$ owns the majority of firm f shares (i.e., $|\underline{N}|\underline{s}_{\text{own}} > 1/2$). Then, Proposition 6 implies that $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$ if $\underline{s}_{\text{other}}/\underline{s}_{\text{own}} < \bar{s}_{\text{other}}/(|\bar{N}|^{-1} + \bar{s}_{\text{own}})$, which holds if group $\bar{N}$ holds sufficiently larger stakes in firm g than group $\underline{N}$.

WAPP models fail to account for how external factors affect the distribution of power among firm f shareholders.²³ As a result, WAPP might overestimate firm f 's coefficient of

²²For $k = n$, adopt the convention $s_{k+1,f} = s_{n+1,f} := 0$. The range of admissible λ_{fg} 's is determined by K for two reasons: (i) Control is exclusive to shareholders (i.e., $s_{if} = 0 \implies \gamma_{if}(s_{*f}) = 0$), and (ii) rank preservation implies that if $s_{if} = s_{jf}$, then $\gamma_{if}(s_{*f}) = \gamma_{jf}(s_{*f})$, so a cut can fall only where the firm f stake strictly drops.

²³Under the specific NB strategic plan that follows the median shareholder's preferences, firm f 's shareholders' stakes in firm g play a role in determining who the median firm f shareholder is. Similar examples can be constructed with other NB models.

effective sympathy towards firm g . The error can be so large that OPM can better explain firm f 's conduct. The econometrician then mistakenly rejects the common ownership hypothesis. Even if WAPP outperforms OPM, the implied $\lambda_{fg}(s)$ may misestimate the true effect of common ownership on firm conduct.²⁴ This will undermine any potential policy implications given that not only the presence but also the *magnitude* of the effect is crucial in determining whether—and what type of—regulatory action is warranted. For example, inflation by WAPP of the estimated impact of common ownership could lead to excessive policy interventions.

The extent of undetected common ownership. Given any ownership structure, $\underline{\lambda} \leq \Lambda_{n_f} = \sum_{i \in N_f(s_{*f})} s_{ig} \leq 1$, so by Proposition 6, for $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$ to hold for both firms f and g , it is necessary that $\max\{\lambda_{fg}^m, \lambda_{gf}^m\} < 1/2$. Proposition 7 shows that this bound is tight. Namely, for any $\lambda^m \in [0, 1/2)$, there exists ownership structure such that $\lambda_{fg}^m = \lambda_{gf}^m = \lambda^m$ and $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$.

To quantify the undetectable potential impact on markups and consumer surplus, consider the case where each firm's marginal cost is c . $\lambda_{fg}^m = \lambda_{gf}^m \equiv \lambda^m$ results in equilibrium markup $\mu_{\text{NB}} := (P_{\text{NB}} - c)/c = (a - c)(1 + \lambda^m)/[(3 + \lambda^m)c]$, consumer surplus $\text{CS}_{\text{NB}} = 2(a - c)^2/[b(3 + \lambda^m)^2]$, and total surplus $\text{TS}_{\text{NB}} = 2(a - c)^2(2 + \lambda^m)/[b(3 + \lambda^m)^2]$. If, instead, each firm maximized its own profit, the equilibrium markup would be $\mu_{\text{OPM}} := (P_{\text{OPM}} - c)/c = (a - c)/(3c)$, consumer surplus would be $\text{CS}_{\text{OPM}} = 2(a - c)^2/(9b)$, and total surplus would be $\text{TS}_{\text{OPM}} = 4(a - c)^2/(9b)$. The percentage increase in the markup and the percentage losses in consumer and total surplus due to common ownership are, respectively, $\mu_{\text{increase}} := (\mu_{\text{NB}} - \mu_{\text{OPM}})/\mu_{\text{OPM}} = 2\lambda^m/(3 + \lambda^m)$, $\text{CS}_{\text{loss}} := (\text{CS}_{\text{OPM}} - \text{CS}_{\text{NB}})/\text{CS}_{\text{OPM}} = 1 - 9/(3 + \lambda^m)^2$, and $\text{TS}_{\text{loss}} := (\text{TS}_{\text{OPM}} - \text{TS}_{\text{NB}})/\text{TS}_{\text{OPM}} = 1 - 9(2 + \lambda^m)/[2(3 + \lambda^m)^2]$, all increasing in λ^m . Substituting in $\lambda^m = 1/2$, we get $\mu_{\text{increase}} = 2/7 \approx 28.6\%$, $\text{CS}_{\text{loss}} = 13/49 \approx 26.5\%$ and $\text{TS}_{\text{loss}} = 4/49 \approx 8.2\%$, arriving at Proposition 7.

Proposition 7. There is an ownership structure at which $\lambda_{fg}^m = \lambda_{gf}^m = \lambda^m$ and OPM explains each firm's conduct better than *any* WAPP and rank-preserving strategic plan (i.e., $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$) if and only if $\lambda^m < 1/2$. Therefore, common ownership can result in up to approximately a 28.6% markup increase, a 26.5% consumer surplus loss, and an 8.2% total surplus loss while still undetected by a test that considers only WAPP models.

Common ownership can increase markups by up to approximately 28.6% and reduce consumer and total surplus by up to approximately 26.5% and 8.2%, respectively, while

²⁴In this example, where there are only two firms and firm f follows the median shareholder's preferences, λ_{fg}^m does not vary with firm g 's quantity. Therefore, rank-preserving WAPP can potentially—namely, when $\lambda_{fg}^m \in [\underline{\lambda}, \bar{\lambda}]$ —perfectly fit the data. Still, it performs worse than OPM when $\underline{\lambda} > 2\lambda_{fg}^m$. However, remember that in general, under NB, the effective $\tilde{\lambda}_{fg}(\alpha_{-f}, s; \pi)$ can vary with external factors, which can never be captured by WAPP. For example, if we simply allow for multiple firms in the example, λ_{fg}^m can vary with competing firms' quantities or costs if $\lambda_{i;f*}$ varies across $i \in N_f(s_{*f})$.

undetected by a test that considers only WAPP models. These numbers are specific to this simple homogeneous product Cournot duopoly example where the firms’ true conduct model follows the median shareholder’s preferences; they are by no means a universal upper bound on the magnitude of the undetectable impact of common ownership. Notice also that the implied range of λ_{fg} is empirically plausible. For example, $\lambda^m = 1/4$ means $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$ if $\underline{\lambda} > 1/2$. Empirical calculations of λ_{fg} tend to exceed that threshold (Backus et al., 2021a).

6 Conclusion

Both theoretical and empirical work has so far followed the WAPP model of firm conduct under common ownership. WAPP assumes that firm f maximizes a weighted average of its shareholders’ portfolio profits from their shares across firms, where the weights depend on firm f ’s ownership structure. A common specification of WAPP assumes proportional control: Each shareholder is assigned weight equal to the number of shares she holds. However, there is limited understanding of the restrictions WAPP imposes on firm conduct. In this paper, I show that WAPP imposes two restrictions: (i) that the firm is efficiently controlled, and (ii) that the distribution of power among shareholders depends only on the firm’s ownership structure, and not on external factors such as the stakes of the firm’s shareholders in competing firms, the strategic choices of other firms, and market conditions (e.g., demand or production technology).

I show that if we drop the second restriction, firm conduct can be represented as the result of asymmetric Nash bargaining (NB) among shareholders. The NB model generalizes WAPP by also requiring efficient control but allowing for external factors to influence the distribution of power among shareholders. I discuss ways in which external factors could play a role in the extent to which each shareholder controls the firm. In addition, I show that the NB model can relax the tension that arises under WAPP between (i) allowing for atomistic shareholders to collectively exert control over the firm and (ii) allowing for large shareholders to have control power. Last, I study the constraints imposed on the parameters of the firm’s objective under WAPP and NB by additional restrictions on firm conduct. Particularly, I characterize firm conduct when “more shares should lead to more control.” I also characterize a popular generalization of WAPP with proportional control used for example by Backus et al. (2021a), Antón et al. (2023), and Ederer and Pellegrino (2025).

Misspecification of the model of firm conduct can undermine tests of the common ownership hypothesis. When external factors influence the distribution of power among shareholders—so that firm conduct is NB but not WAPP—and an econometrician considers only WAPP models, I show that own profit maximization may fit firm conduct better than *any* specification of WAPP, even though common ownership affects firm conduct. I

conclude that failing to account for how external factors affect the distribution of power among shareholders can lead to a false rejection of the common ownership hypothesis and a failure to detect large effects of common ownership on markups and consumer surplus.

The results guide researchers and practitioners to think in two steps when deciding on a model of firm conduct under common ownership: (i) Choose between WAPP and NB depending on whether the assumption that the distribution of power among the firm’s shareholders is independent of external factors is likely to be satisfied or not, and (ii) decide on a specific parametrization of the model chosen in the first step depending on what additional conditions are likely satisfied.

The results also outline a clear path for future research. Future theoretical and empirical work can evaluate the robustness of results obtained under WAPP by also considering NB models of firm conduct.²⁵ For example, Backus et al. (2021b) find that own profit maximization is more consistent with firm conduct in the ready-to-eat cereal market than WAPP with proportional control, which may serve as evidence against the common ownership hypothesis. A natural robustness check is to consider different parametrizations of WAPP. For instance, one can use the generalization of proportional control considered by Backus et al. (2021a), Antón et al. (2023), Ederer and Pellegrino (2025), the suitability of which can be judged based on the characterization provided in this paper. However, if one is concerned that the distribution of power among shareholders may depend on external factors, this would not be enough. One would have to test NB models against own profit maximization to more robustly evaluate the common ownership hypothesis. Doing so could also help explain why several studies find common ownership to affect firm conduct while some find limited evidence to that effect: Misspecification of the model of firm conduct under common ownership could sometimes lead to false negatives, where researchers fail to reject the null hypothesis that common ownership does not affect firm conduct when it actually does.

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²⁵The Online Appendix studies a simple duopoly comparing two scenarios: (i) both firms’ strategic plans are WAPP with proportional control or (ii) both firms’ strategic plans are NB with random dictatorship disagreement payoffs and weights proportional to shares. It shows that theoretical predictions and policy implications can differ significantly between the two scenarios.

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Appendix

A Proofs

Proof of Proposition 1. Part (i). Let R_f be WAPP with control power function γ_{*f} . For every $s \in S$ define $\widetilde{N}(s_{*f}) := \{i \in N : \gamma_{if}(s_{*f}) > 0\}$, and use $\widetilde{N}(s_{*f})$ to verify that R_f satisfies the strong efficiency conditions.

Part (ii). Now, assume R_f is NB with bargaining power function β_{*f} . For every $s \in S$ define $\widetilde{N}(s_{*f}) := \{i \in N : \beta_{if}(s_{*f}) > 0\}$, and use $\widetilde{N}(s_{*f})$ to verify that R_f satisfies the weak efficiency conditions.

Part (iii). Let R_f be NB with strict benefits from agreement and bargaining power function β_{*f} . For every $s \in S$ define $\widetilde{N}(s_{*f}) := \{i \in N : \beta_{if}(s_{*f}) > 0\}$, and use $\widetilde{N}(s_{*f})$ to verify that R_f satisfies the strong efficiency conditions. Also, since for every s , α_{-f} , and π , there exists $u \in \mathcal{U}_f(\alpha_{-f}, s; \pi)$ such that $u_i > d_{if}(\alpha_{-f}, s; \pi)$ for every $i \in N_f(\beta_{*f}(s_{*f}))$, the Nash product $\prod_{i \in N_f(\beta_{*f}(s_{*f}))} (u_i - d_{if})^{\beta_{if}}$ is strictly quasiconcave in $(u_i)_{i \in N_f(\beta_{*f}(s_{*f}))}$ where that inequality holds. Thus, since $\{u \in \mathcal{U}_f(\alpha_{-f}, s; \pi) : u_i > d_{if}(\alpha_{-f}, s; \pi) \text{ for every } i \in N_f(\beta_{*f}(s_{*f}))\}$ is convex for every $s \in S$, $\pi \in \Pi$, and $\alpha_{-f} \in \times_{g \neq f} \Delta(A_g)$, there exists at most one $(u_i(\alpha_f, \alpha_{-f}, s_{i*}))_{i \in N_f(\beta_{*f}(s_{*f}))}$ that maximizes the Nash product, which is condition (iv) of Definition 3; condition (v) holds because the Nash product and $B_f(\alpha_{-f}, s; \pi)$ depend on α_f only through $(u_i(\alpha_f, \alpha_{-f}, s_{i*}))_{i \in N_f(\beta_{*f}(s_{*f}))}$, so R_f is internally consistent.

Part (iv). Let R_f be strongly efficient and internally consistent, so that there exists function $\widetilde{N}(s_{*f})$ satisfying the conditions of Definition 3. For every $s \in S$, let the bargaining power function be

$$\beta_{*f}(s_{*f}) := \frac{1}{|\widetilde{N}(s_{*f})|} \left(\mathbb{I}(1 \in \widetilde{N}(s_{*f})) \quad \dots \quad \mathbb{I}(n \in \widetilde{N}(s_{*f})) \right),$$

where $\mathbb{I}$ is the indicator function, and for every $i \in \widetilde{N}(s_{*f})$, let the disagreement payoff function be $d_{if}(\alpha_{-f}, s; \pi) := u_i(\widetilde{\alpha}_f(\alpha_{-f}, s; \pi), \alpha_{-f}, s_{i*})$ for some function $\widetilde{\alpha}_f(\alpha_{-f}, s; \pi)$ that is a selection from $R_f(\alpha_{-f}, s; \pi)$ (i.e., $\widetilde{\alpha}_f(\alpha_{-f}, s; \pi) \in R_f(\alpha_{-f}, s; \pi)$), and for every $i \notin \widetilde{N}(s_{*f})$, let $d_{if}(\alpha_{-f}, s; \pi) := 0$. d_{*f} is well-defined since R_f is internally consistent. Notice that by the way β_{*f} is defined, $N_f(\beta_{*f}(s_{*f})) = \widetilde{N}(s_{*f})$. Fix arbitrary $s \in S$, $\pi \in \Pi$, and $\alpha_{-f} \in \times_{g \neq f} \Delta(A_g)$. Observe that any $\alpha_f \in R_f(\alpha_{-f}, s; \pi)$ achieves the maximum value of zero for the Nash product, so

$$R_f(\alpha_{-f}, s; \pi) \subseteq \arg \max_{\alpha'_f \in B_f(\alpha_{-f}, s; \pi)} \left\{ \prod_{i \in N_f(\beta_{*f}(s_{*f}))} \left(u_i(\alpha'_f, \alpha_{-f}, s_{i*}) - d_{if}(\alpha_{-f}, s; \pi) \right)^{\beta_{if}(s_{*f})} \right\}.$$

Now, take an arbitrary

$$\alpha_f \in \arg \max_{\alpha'_f \in B_f(\alpha_{-f}, s; \pi)} \left\{ \prod_{i \in N_f(\beta_{*f}(s_{*f}))} \left(u_i(\alpha'_f, \alpha_{-f}, s_{i*}) - d_{if}(\alpha_{-f}, s; \pi) \right)^{\beta_{if}(s_{*f})} \right\}.$$

We will show by contradiction that $\alpha_f \in R_f(\alpha_{-f}, s; \pi)$. Assume that $\alpha_f \notin R_f(\alpha_{-f}, s; \pi)$. Since R_f is internally consistent, there exists $j \in N_f(\beta_{*f}(s_{*f}))$ such that $u_j(\alpha_f, \alpha_{-f}, s_{j*}) \neq u_j(\tilde{\alpha}_f(\alpha_{-f}, s; \pi), \alpha_{-f}, s_{j*}) = d_{jf}(\alpha_{-f}, s; \pi)$. Also, given that α_f maximizes the Nash product above, $\alpha_f \in B_f(\alpha_{-f}, s; \pi)$, and thus, $u_i(\alpha_f, \alpha_{-f}, s_{i*}) \geq d_{if}(\alpha_{-f}, s; \pi)$ for every $i \in N_f(\beta_{*f}(s_{*f}))$. Particularly, the inequality must hold strictly for j , that is, $u_j(\alpha_f, \alpha_{-f}, s_{j*}) > d_{jf}(\alpha_{-f}, s; \pi) \equiv u_j(\tilde{\alpha}_f(\alpha_{-f}, s; \pi), \alpha_{-f}, s_{j*})$. But then, α_f weakly dominates $\tilde{\alpha}_f(\alpha_{-f}, s; \pi) \in R_f(\alpha_{-f}, s; \pi)$, a contradiction to the strong efficiency of R_f . Therefore,

$$R_f(\alpha_{-f}, s; \pi) \supseteq \arg \max_{\alpha_f \in B_f(\alpha_{-f}, s; \pi)} \left\{ \prod_{i \in N_f(\beta_{*f}(s_{*f}))} \left(u_i(\alpha_f, \alpha_{-f}, s_{i*}) - d_{if}(\alpha_{-f}, s; \pi) \right)^{\beta_{if}(s_{*f})} \right\}.$$

Part (v). Let R_f be WAPP with control power function γ_{*f} , and for every $s \in S$ define $\tilde{N}(s_{*f}) := \{i \in N : \gamma_{if}(s_{*f}) > 0\} \subseteq N_f(s_{*f})$. Given part (i), it remains to show conditions (iv) and (v) of Definition 3 hold. Condition (v) holds because the WAPP objective depends on α_f only through $(u_i(\alpha_f, \alpha_{-f}, s_{i*}))_{i \in \tilde{N}(s_{*f})}$. Also, the WAPP objective is strictly concave in a_f as a convex combination of $(u_i(a_f, \alpha_{-f}, s_{i*}))_{i \in \tilde{N}(s_{*f})}$, which are strictly concave in a_f , so $R_f(\alpha_{-f}, s; \pi)$ is a singleton, and thus condition (iv) holds.²⁶

Now, let R_f be NB with bargaining power function β_{*f} and disagreement payoff function d_{*f} , and for every $s \in S$, define $\emptyset \neq \tilde{N}(s_{*f}) := N_f(\beta_{*f}(s_{*f})) \subseteq N_f(s_{*f})$ by condition (ii) of the definition of NB, so conditions (i) and (ii) of Definition 3 hold. Fix s , α_{-f} and π ; write, in what follows, $u_i(\cdot)$ for $u_i(\cdot, \alpha_{-f}, s_{i*})$ and d_i for $d_{if}(\alpha_{-f}, s; \pi)$, B_f for $B_f(\alpha_{-f}, s; \pi)$, and define $\Phi(\alpha_f) := \prod_{i \in \tilde{N}(s_{*f})} (u_i(\alpha_f) - d_i)^{\beta_{if}(s_{*f})}$, so that $R_f(\alpha_{-f}, s; \pi) = \arg \max_{\alpha_f \in B_f} \Phi(\alpha_f)$. Since for every α_f , whether $\alpha_f \in B_f$ and the value of Φ depend on α_f only through $(u_i(\alpha_f))_{i \in \tilde{N}(s_{*f})}$, condition (v) of Definition 3 holds. From strict concavity of $(u_i(a_f, \alpha_{-f}, s_{i*}))_{i \in \tilde{N}(s_{*f})}$ in a_f , it also follows that $R_f(\alpha_{-f}, s; \pi)$ is a singleton, so condition (iv) of Definition 3 holds. Last, take any $\alpha_f \in R_f(\alpha_{-f}, s; \pi)$ and $\alpha''_f \in \Delta(A_f)$ such that $u_i(\alpha''_f) \geq u_i(\alpha_f)$ for every $i \in \tilde{N}(s_{*f})$. Then, $\Phi(\alpha''_f) \geq \Phi(\alpha_f)$, so $\alpha''_f \in R_f(\alpha_{-f}, s; \pi)$ and given that condition (iv) of Definition 3 holds, $u_i(\alpha''_f) = u_i(\alpha_f)$. Thus, condition (iii) of Definition 3 holds. **Q.E.D.**

Proof of Proposition 2. Part (i). Let R_f be GWAPP with control power function γ_{*f} . Fix some α_{-f} , $(s_{*g})_{g \neq f}$, and π . For every s_{*f} , define $\tilde{N}(s_{*f}) := \{i \in N : \gamma_{if}(\alpha_{-f}, s; \pi) > 0\}$.

²⁶Notice that for every non-degenerate $\alpha_f \in \Delta(A_f)$, $\bar{a}_f := \sum_{k=1}^{\ell} \alpha_f(a_f^k) a_f^k \in A_f$ and satisfies $u_i(\bar{a}_f, \alpha_{-f}, s_{i*}) > u_i(\alpha_f, \alpha_{-f}, s_{i*})$ for every $i \in N_f(s_{*f})$.

By condition (iii) of Definition 4, $\widetilde{N}(s_{*f})$ is independent of α_{-f} , $(s_{*g})_{g \neq f}$, and π . Use $\widetilde{N}(s_{*f})$ to verify that R_f satisfies the strong efficiency conditions.

Part (ii)(a). Assume R_f is GWAPP. Then, by part (i) of this Proposition, it is strongly efficient, and, as for the case of WAPP in the proof of part (v) of Proposition 1, R_f is also internally consistent. Part (iv) of Proposition 1 then implies that R_f is NB.

Part (ii)(b). Assume that R_f is NB with strict benefits from agreement. By Proposition 1, it is strongly efficient and internally consistent with $\widetilde{N}(s_{*f}) = N_f(\beta_{*f}(s_{*f})) \subseteq N_f(s_{*f})$. Take some arbitrary α_{-f} , s , and π write $d_i := d_{if}(\alpha_{-f}, s; \pi)$ for every $i \in N$, and let $v^* := (u_i(\alpha_f, \alpha_{-f}, s_{i*}))_{i \in \widetilde{N}(s_{*f})}$ for every $\alpha_f \in R_f(\alpha_{-f}, s; \pi)$, which is well-defined by internal consistency.

The logarithm $L(v) := \sum_{i \in \widetilde{N}(s_{*f})} \beta_{if}(s_{*f}) \ln(v_i - d_i)$ of the Nash product has $\partial L(v^*) / \partial v_i = K \tilde{\gamma}_{if}(\alpha_{-f}, s; \pi)$ for every $i \in \widetilde{N}(s_{*f})$, where $K := \sum_{j \in \widetilde{N}(s_{*f})} \beta_{jf}(s_{*f}) / (v_j^* - d_j) > 0$. For arbitrary $\alpha_f \in \Delta(A_f)$, let $v := (u_i(\alpha_f, \alpha_{-f}, s_{i*}))_{i \in \widetilde{N}(s_{*f})}$, and define $\phi(t) := L((1-t)v^* + tv)$. Because $v_i^* > d_i$ for every $i \in \widetilde{N}(s_{*f})$, $\phi(t)$ is well-defined and differentiable at $t = 0$. Thus, since v^* maximizes the Nash product, $\phi'(0) = K \sum_{i \in \widetilde{N}(s_{*f})} \tilde{\gamma}_{if}(\alpha_{-f}, s; \pi)(v_i - v_i^*) \leq 0$, so $\sum_{i \in \widetilde{N}(s_{*f})} \tilde{\gamma}_{if}(\alpha_{-f}, s; \pi)(v_i - v_i^*) \leq 0$. As α_f was arbitrary,

$$R_f(\alpha_{-f}, s; \pi) \subseteq \arg \max_{\alpha_f \in \Delta(A_f)} \left\{ \sum_{i \in \widetilde{N}(s_{*f})} \tilde{\gamma}_{if}(\alpha_{-f}, s; \pi) u_i(\alpha_f, \alpha_{-f}, s_{i*}) \right\}. \quad (1)$$

As in the proof of part (v) of Proposition 1, it follows that the right-hand side of (1) is a singleton, so (1) holds with equality. We conclude that R_f is GWAPP with $\gamma_{*f}(\alpha_{-f}, s; \pi) := \tilde{\gamma}_{*f}(\alpha_{-f}, s; \pi)$ for every $i \in \widetilde{N}(s_{*f})$, well-defined because $v_i^* > d_i$ for every $i \in \widetilde{N}(s_{*f})$, and $\gamma_{*f}(\alpha_{-f}, s; \pi) := 0$ for every $i \notin \widetilde{N}(s_{*f})$. Condition (iii) of Definition 4 holds because $N_f(\gamma_{*f}(\alpha_{-f}, s; \pi)) = \widetilde{N}(s_{*f}) = N_f(\beta_{*f}(s_{*f}))$ depends only on s_{*f} . **Q.E.D.**

Proof of Proposition 3. (i) Firm f 's objective function is $\sum_{k \in N} \gamma_{kf}(s_{*f}) u_k(a_f, a_{-f}, s_{k*})$, and the Implicit Function Theorem gives that around any f -unanimous s

$$\begin{aligned} \nabla_{ds} R_f(a_{-f}, s) &= - \frac{(\gamma_{if}(s_{*f}) - \gamma_{jf}(s_{*f})) \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \Big|_{a_f = R_f(a_{-f}, s)}}{\sum_{k \in N} \gamma_{kf}(s_{*f}) \frac{\partial^2 u_k(a_f, a_{-f}, s_{k*})}{\partial a_f^2} \Big|_{a_f = R_f(a_{-f}, s)}} \implies \\ \text{sgn} \{ \nabla_{ds} R_f(a_{-f}, s) \} &= \text{sgn} \left\{ (\gamma_{if}(s_{*f}) - \gamma_{jf}(s_{*f})) \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \Big|_{a_f = R_f(a_{-f}, s)} \right\}. \end{aligned} \quad (2)$$

We will prove each direction separately.

$\Leftarrow$: Assume that for every s_{*f} and every pair of firm f shareholders $i, j \in N_f(s_{*f})$, $s_{if} \geq s_{jf} \implies \gamma_{if}(s_{*f}) \geq \gamma_{jf}(s_{*f})$. We need to show that R_f is rank-preserving. Take

arbitrary $a_{-f} \in A_{-f}$, f -unanimous $s \in S$, and pair of distinct shareholders $i, j \in N_f(s_{*f})$ with $s_{if} \geq s_{jf}$. It follows that $\gamma_{if}(s_{*f}) \geq \gamma_{jf}(s_{*f})$. Consider a stock trade where i buys firm $g \neq f$ shares from j . (2) gives

$$\left. \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \right|_{a_f=R_f(a_{-f}, s)} \stackrel{(\text{resp. } \leq)}{\geq} 0 \implies \nabla_{ds} R_f(a_{-f}, s) \stackrel{(\text{resp. } \leq)}{\geq} 0.$$

$\implies$: Let R_f be rank-preserving. We need to show that for every s_{*f} and every pair of firm f shareholders $i, j \in N_f(s_{*f})$, $s_{if} \geq s_{jf} \implies \gamma_{if}(s_{*f}) \geq \gamma_{jf}(s_{*f})$. Take arbitrary s_{*f} and pair of firm f shareholders $i, j \in N_f(s_{*f})$ with $s_{if} \geq s_{jf}$. By assumption, there exist $(s_{*g})_{g \neq f}$, firm $g \neq f$, and $a_{-f} \in A_{-f}$ such that s is f -unanimous and $\partial \pi_g(a_f, a_{-f}) / \partial a_f|_{a_f=R_f(a_{-f}, s)} \neq 0$.²⁷ Let $ds := (\mathbf{e}_i - \mathbf{e}_j) \otimes \mathbf{e}_g$. If $\partial \pi_g(a_f, a_{-f}) / \partial a_f|_{a_f=R_f(a_{-f}, s)} > 0$, then given that R_f is rank-preserving, $\nabla_{ds} R_f(a_{-f}, s) \geq 0$, and thus (2) implies $\gamma_{if}(s_{*f}) \geq \gamma_{jf}(s_{*f})$. Similarly, if $\partial \pi_g(a_f, a_{-f}) / \partial a_f|_{a_f=R_f(a_{-f}, s)} < 0$, then given that R_f is rank-preserving, $\nabla_{ds} R_f(a_{-f}, s) \leq 0$, and thus (2) implies $\gamma_{if}(s_{*f}) \geq \gamma_{jf}(s_{*f})$.

(ii) Under NB, for any f -unanimous s , $\partial u_k(a_f, a_{-f}, s_{k*}) / \partial a_f|_{a_f=R_f(a_{-f}, s)} = 0$ for every shareholder k of firm f . For a stock trade ds , the Implicit Function Theorem then gives that around any f -unanimous s ²⁸

$$\begin{aligned} \text{sgn} \{ \nabla_{ds} R_f(a_{-f}, s) \} &= \text{sgn} \left\{ (\tilde{\gamma}_{if}(a_{-f}, s) - \tilde{\gamma}_{jf}(a_{-f}, s)) \left. \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \right|_{a_f=R_f(a_{-f}, s)} \right\} \\ &= \text{sgn} \left\{ \frac{(\beta_{if}(s_{*f}) - \beta_{jf}(s_{*f}) s_{if} / s_{jf}) \left. \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \right|_{a_f=R_f(a_{-f}, s)}}{u_i(R_f(a_{-f}, s), a_{-f}, s_{i*}) - u_i(\alpha_d(a_{-f}, s), a_{-f}, s_{i*})} \right\} \\ &= \text{sgn} \left\{ \left(\frac{\beta_{if}(s_{*f})}{s_{if}} - \frac{\beta_{jf}(s_{*f})}{s_{jf}} \right) \left. \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \right|_{a_f=R_f(a_{-f}, s)} \right\}, \quad (3) \end{aligned}$$

where $\alpha_d(a_{-f}, s)$ the strategy followed in case of disagreement. In the second line, we have used the fact that $\lambda_{i;f*} = \lambda_{j;f*}$, and in the third the fact that $u_i(R_f(a_{-f}, s), a_{-f}, s_{i*}) > u_i(\alpha_d(a_{-f}, s), a_{-f}, s_{i*})$. We will prove each direction separately.

$\Leftarrow$: Assume that for every s_{*f} and every pair of firm f shareholders $i, j \in N_f(s_{*f})$, $s_{if} \geq s_{jf} \implies \beta_{if}(s_{*f}) / s_{if} \geq \beta_{jf}(s_{*f}) / s_{jf}$. We need to show that R_f is rank-preserving. Take arbitrary $a_{-f} \in A_{-f}$, f -unanimous $s \in S$, and pair of distinct shareholders $i, j \in N_f(s_{*f})$ with $s_{if} \geq s_{jf}$. It follows that $\beta_{if}(s_{*f}) / s_{if} \geq \beta_{jf}(s_{*f}) / s_{jf}$. Consider a stock trade where i buys firm $g \neq f$ shares from j . (3) gives

$$\left. \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \right|_{a_f=R_f(a_{-f}, s)} \stackrel{(\text{resp. } \leq)}{\geq} 0 \implies \nabla_{ds} R_f(a_{-f}, s) \stackrel{(\text{resp. } \leq)}{\geq} 0.$$

²⁷For example, $(s_{*g})_{g \neq f}$ such that $s_{*g} = s_{*f}$ for every $g \neq f$ makes s f -unanimous.

²⁸Notice that because $\partial u_k(a_f, a_{-f}, s_{k*}) / \partial a_f|_{a_f=R_f(a_{-f}, s)} = 0$ for every shareholder k of firm f , the changes in $\tilde{\gamma}_{*f}$ caused by the stock trade vanish.

$\implies$: Let R_f be NB and rank-preserving. We need to show that for every s_{*f} and every pair of firm f shareholders $i, j \in N_f(s_{*f})$, $s_{if} \geq s_{jf} \implies \beta_{if}(s_{*f})/s_{if} \geq \beta_{jf}(s_{*f})/s_{jf}$. Take arbitrary s_{*f} and pair of firm f shareholders $i, j \in N_f(s_{*f})$ with $s_{if} \geq s_{jf}$. By assumption, there exist $(s_g)_{g \neq f}$, firm $g \neq f$, and $a_{-f} \in A_{-f}$ such that s is f -unanimous and $\partial \pi_g(a_f, \hat{a}_{-f})/\partial a_f|_{a_f=R_f(a_{-f}, s)} \neq 0$. Let $ds := (\mathbf{e}_i - \mathbf{e}_j) \otimes \mathbf{e}_g$. If $\partial \pi_g(a_f, a_{-f})/\partial a_f|_{a_f=R_f(a_{-f}, s)} > 0$, then given that R_f is rank-preserving, $\nabla_{ds} R_f(a_{-f}, s) \geq 0$, and thus (3) implies $\beta_{if}(s_{*f})/s_{if} \geq \beta_{jf}(s_{*f})/s_{jf}$. If $\partial \pi_g(a_f, a_{-f})/\partial a_f|_{a_f=R_f(a_{-f}, s)} < 0$, then given that R_f is rank-preserving, $\nabla_{ds} R_f(a_{-f}, s) \leq 0$, and thus (3) implies $\beta_{if}(s_{*f})/s_{if} \geq \beta_{jf}(s_{*f})/s_{jf}$. **Q.E.D.**

Proof of Lemma 1. (i-a) Firm f 's objective function is $\sum_{k \in N_f(\gamma_{*f}(s'_{*f}))} \gamma_{kf}(s_{*f}) u_k(a_f, a_{-f}, s_{k*})$, and the Implicit Function Theorem gives that around any f -unanimous s ,

$$\begin{aligned} & \text{sgn} \{ \nabla_{ds} R_f(a_{-f}, s) \} \\ &= \text{sgn} \left\{ \left((1 - \psi) \gamma_{if}(s_{*f}) - \psi \sum_{j \in N_f(s_{*f}) \setminus \{i\}} \gamma_{jf}(s_{*f}) \right) \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \Big|_{a_f=R_f(a_{-f}, s)} \right\}. \end{aligned} \quad (4)$$

We will prove each direction separately.

$\Leftarrow$: Let R_f be WAPP, and assume that $\gamma_{if}(s_{*f}) = \psi$. We need to show that the stock trade is neutral. Take arbitrary $a_{-f} \in A_{-f}$. (4) then gives

$$\text{sgn} \{ \nabla_{ds} R_f(a_{-f}, s) \} = \text{sgn} \left\{ ((1 - \psi)\psi - \psi(1 - \psi)) \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \Big|_{a_f=R_f(a_{-f}, s)} \right\} = 0.$$

$\implies$: Let R_f be WAPP, and assume that the stock trade is neutral. We need to show that $\gamma_{if}(s_{*f}) = \psi$. By assumption, there exists $a_{-f} \in A_{-f}$ such that, evaluated at $a_f = R_f(a_{-f}, s)$, $\partial \pi_g(a_f, a_{-f})/\partial a_f \neq 0$. Then, (4) implies $\gamma_{if}(s_{*f}) = \psi$.

(ii-a) Under NB, the Implicit Function Theorem gives that around any f -unanimous s

$$\begin{aligned} & \text{sgn} \{ \nabla_{ds} R_f(a_{-f}, s) \} \\ &= \text{sgn} \left\{ \left((1 - \psi) \tilde{\gamma}_{if}(a_{-f}, s) - \psi \sum_{j \in N_f(s_{*f}) \setminus \{i\}} \tilde{\gamma}_{jf}(a_{-f}, s) \right) \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \Big|_{a_f=R_f(a_{-f}, s)} \right\} \\ &= \text{sgn} \left\{ \left((1 - \psi) \frac{\beta_{if}(s_{*f})}{s_{if}} - \psi \sum_{j \in N_f(s_{*f}) \setminus \{i\}} \frac{\beta_{jf}(s_{*f})}{s_{jf}} \right) \frac{\partial \pi_g(a_f, a_{-f})}{\partial a_f} \Big|_{a_f=R_f(a_{-f}, s)} \right\}, \end{aligned}$$

where the second line follows as in the proof of Proposition 3. Then, the result follows as in part (i).

Similar arguments prove (i-b) and (ii-b).

Q.E.D.

Proof of Proposition 4. (i) $\Rightarrow$: Assume that R_f has stock-trade monotone control, and take arbitrary s_{*f} , pair of firm f shareholders $i, j \in N_f(s_{*f})$, and $s'_{*f} = s_{*f} + t(\mathbf{e}_i - \mathbf{e}_j)$ for some $t \in [0, \min\{s_{jf}, 1 - s_{if}\}]$. There exist $(s_{*g})_{g \neq f}$ and $(s'_{*g})_{g \neq f}$ such that s and s' are f -unanimous ownership matrices with $s'_{i*} = (1 + t/s_{if})s_{i*}$ and $s'_{j*} = s_{j*} - (t/s_{if})s_{i*}$. Given Lemma 1, starting from s , a $(\gamma_{if}(s_{*f}), g, i, N_f(s_{*f}) \setminus \{i\})$ -stock trade is f -neutral. Also, starting from s' , a $(\gamma_{if}(s'_{*f}), g, i, N_f(s'_{*f}) \setminus \{i\})$ -stock trade is f -neutral. Thus, given that R_f has monotone control power, $\gamma_{if}(s'_{*f}) \geq \gamma_{if}(s_{*f})$.

$\Leftarrow$: Assume that for every s_{*f} , every pair of firm f shareholders $i, j \in N_f(s_{*f})$, and $t \in [0, \min\{s_{jf}, 1 - s_{if}\}]$, $\gamma_{if}(s_{*f} + t(\mathbf{e}_i - \mathbf{e}_j)) \geq \gamma_{if}(s_{*f})$. Now, take arbitrary $g \neq f$, pair of shareholders $i, j \in N$, and f -unanimous ownership matrix s . Assume that starting from s , a $(\psi, g, i, N_f(s_{*f}) \setminus \{i\})$ -stock trade is f -neutral, and that starting from s' such that $s'_{k*} = s_{k*}$ for every $k \neq i, j$, $s'_{i*} = (1 + t/s_{if})s_{i*}$, and $s'_{j*} = s_{j*} - (t/s_{if})s_{i*}$ for some $t \in [0, \min_{g \in M: s_{ig} > 0} \min\{1 - s_{ig}/s_{ig}, s_{jg}/s_{ig}\}]$, a $(\psi', g, i, N_f(s'_{*f}) \setminus \{i\})$ -stock trade is f -neutral. s' is f -unanimous given that s is. Then, given Lemma 1, $\psi = \gamma_{if}(s_{*f})$ and $\psi' = \gamma_{if}(s'_{*f})$, where $s'_{*f} = s_{*f} + t(\mathbf{e}_i - \mathbf{e}_j)$. Therefore, $\psi' \geq \psi$.

Part (ii) follows similarly.

Q.E.D.

Proof of Proposition 5. (i) $\Leftarrow$: Assume that there exists $\delta : [0, 1) \rightarrow \mathbb{R}_+$ with $\delta(0) = 0$ and $\delta(x) > 0$ for every $x > 0$ such that for every s and $i \in N$, $\gamma_{if}(s_{*f}) = \delta(s_{if}) / \sum_{j \in N} \delta(s_{jf})$. The strategic plan R_f is anonymous. To see why it is inclusive, take arbitrary firm $g \neq f$, f -unanimous ownership matrix s , and pair of firm f shareholders $i, j \in N_f(s_{*f})$.

It remains to show that R_f satisfies IIS. Take arbitrary $g \neq f$, f -unanimous ownership matrices s and s' , pair of shareholders $i, j \in N_f(s_{*f})$, and $\psi \in [0, 1]$ such that $s'_{if} = s_{if}$ and $s'_{jf} = s_{jf}$. Take any (ψ, g, i, j) -stock trade that is f -neutral starting from s . Lemma 1 implies $\gamma_{if}(s_{*f})(1 - \psi) = \gamma_{jf}(s_{*f})\psi$. Multiplying both sides by $\gamma_{jf}(s'_{*f})$, we get $\gamma_{if}(s_{*f})\gamma_{jf}(s'_{*f})(1 - \psi) = \gamma_{jf}(s_{*f})\gamma_{jf}(s'_{*f})\psi$. Substituting $\gamma_{if}(s_{*f})\gamma_{jf}(s'_{*f}) = \gamma_{if}(s'_{*f})\gamma_{jf}(s_{*f})$ (which follows from $\gamma_{if}(s_{*f}) = \delta(s_{if}) / \sum_{j \in N} \delta(s_{jf})$, $s'_{if} = s_{if}$, and $s'_{jf} = s_{jf}$) in the left-hand side, we get $\gamma_{if}(s'_{*f})\gamma_{jf}(s_{*f})(1 - \psi) = \gamma_{jf}(s_{*f})\gamma_{jf}(s'_{*f})\psi$. Given that $\gamma_{jf}(s_{*f}) = \delta(s_{jf}) / \sum_{k \in N} \delta(s_{kf}) > 0$, we can divide both sides by $\gamma_{jf}(s_{*f})$, which gives $\gamma_{if}(s'_{*f})(1 - \psi) = \gamma_{jf}(s'_{*f})\psi$. Therefore, given Lemma 1, the (ψ, g, i, j) -stock trade is f -neutral also starting from s' .

$\Rightarrow$: Assume that R_f satisfies anonymity, inclusivity, and IIS. That R_f is WAPP implies that there exists $\gamma_{*f} : \Delta^n \rightarrow \Delta^n$ satisfying the conditions of Definition 1.

Now, notice that for any s_{*f} and $i \in N_f(s_{*f})$, $\gamma_{if}(s_{*f}) > 0$. To see this, take arbitrary s_{*f} and $i \in N_f(s_{*f})$. There exist $(s_{*g})_{g \neq f}$ such that s is f -unanimous. By inclusivity, for any firm $g \neq f$ and shareholder $j \in N_f(s_{*f}) \setminus \{i\}$, a $(0, g, i, j)$ -stock trade is not f -neutral, which by Lemma 1 means that $\gamma_{if}(s_{*f}) \neq 0$, so $\gamma_{if}(s_{*f}) > 0$.

Now, take arbitrary s_{*f} , s'_{*f} , and pair of shareholders $i, j \in N_f(s_{*f})$ such that $s'_{if} = s_{if}$ and $s'_{jf} = s_{jf}$. There exist $(s_{*g})_{g \neq f}$ and $(s'_{*g})_{g \neq f}$ such that s and s' are f -unanimous

ownership matrices. Since R_f satisfies IIS, for any $\psi \in [0,1]$, if starting from s , a (ψ, g, i, j) -stock trade is f -neutral, then starting from s' , a (ψ, g, i, j) -stock trade is again f -neutral. Given Lemma 1, (i) starting from s , a (ψ, g, i, j) -stock trade is f -neutral if and only if $\psi = \gamma_{if}(s_{*f})/(\gamma_{if}(s_{*f}) + \gamma_{jf}(s_{*f}))$, and (ii) starting from s' , a (ψ', g, i, j) -stock trade is f -neutral if and only if $\psi' = \gamma_{if}(s'_{*f})/(\gamma_{if}(s'_{*f}) + \gamma_{jf}(s'_{*f}))$. Thus, $\gamma_{if}(s_{*f})/(\gamma_{if}(s_{*f}) + \gamma_{jf}(s_{*f})) = \gamma_{if}(s'_{*f})/(\gamma_{if}(s'_{*f}) + \gamma_{jf}(s'_{*f}))$, which implies $\gamma_{if}(s_{*f})/\gamma_{jf}(s_{*f}) = \gamma_{if}(s'_{*f})/\gamma_{jf}(s'_{*f})$. This means that $\gamma_{if}(s_{*f})/\gamma_{jf}(s_{*f})$ depends on s_{*f} only through s_{if} and s_{jf} . Therefore, there exists function $h_{ij} : \{(x, y) \in \mathbb{R}_{++}^2 : x + y < 1\} \rightarrow \mathbb{R}_{++}$ such that $\gamma_{if}(s_{*f}) = h_{ij}(s_{if}, s_{jf})\gamma_{jf}(s_{*f})$ for every s_{*f} . Anonymity implies $\nabla_{ds}R_f(a_{-f}, s) = \nabla_{Pds}R_f(a_{-f}, Ps)$, so for any permutation matrix P with $Pe_i = e_{i'}$ and $Pe_j = e_{j'}$, a (ψ, g, i, j) -stock trade is f -neutral starting from s if and only if a (ψ, g, i', j') -stock trade is f -neutral starting from Ps , which is f -unanimous with $(Ps)_{i'f} = s_{if}$ and $(Ps)_{j'f} = s_{jf}$; Lemma 1 then gives $h_{i'j'} = h_{ij}$, so we can drop the subscript ij from h ; namely, there exists $h : \{(x, y) \in \mathbb{R}_{++}^2 : x + y < 1\} \rightarrow \mathbb{R}_{++}$ such that $h(s_{if}, s_{jf}) = \gamma_{if}(s_{*f})/\gamma_{jf}(s_{*f})$ for every s_{*f} and every $i, j \in N_f(s_{*f})$. Notice that for every s_{*f} and every $i, j, k \in N_f(s_{*f})$

$$h(s_{if}, s_{jf}) = \frac{\gamma_{if}(s_{*f})}{\gamma_{jf}(s_{*f})} = \frac{\gamma_{if}(s_{*f})/\gamma_{kf}(s_{*f})}{\gamma_{jf}(s_{*f})/\gamma_{kf}(s_{*f})} = \frac{h(s_{if}, s_{kf})}{h(s_{jf}, s_{kf})}.$$

This means that for every $x, y, z > 0$ such that $x + y + z \leq 1$, $h(x, y) = h(x, z)/h(y, z)$. In fact, this equation must hold more generally. To see this, take arbitrary $x, y, z > 0$ such that $x + y < 1$, $x + z < 1$ and $y + z < 1$. It then holds that

$$\begin{aligned} h(x, y) &= \frac{h(x, 1 - \max\{x + y, x + z, y + z\})}{h(y, 1 - \max\{x + y, x + z, y + z\})} \\ &= \frac{h(x, 1 - \max\{x + y, x + z, y + z\})/h(z, 1 - \max\{x + y, x + z, y + z\})}{h(y, 1 - \max\{x + y, x + z, y + z\})/h(z, 1 - \max\{x + y, x + z, y + z\})} \\ &= \frac{h(x, z)}{h(y, z)}, \end{aligned} \tag{5}$$

where the first and third lines follow from $h(x, y) = h(x, z)/h(y, z)$ holding for every $x, y, z > 0$ such that $x + y + z \leq 1$.

Now, define $\delta : [0, 1] \rightarrow \mathbb{R}_+$ given by

$$\delta(x) := \begin{cases} 0 & \text{if } x = 0 \\ h(x, 1/5) & \text{if } x \in (0, 3/4] \\ \frac{h(x, (1-x)/5)}{h(1/5, (1-x)/5)} & \text{if } x \in (3/4, 1) \end{cases}$$

and satisfying $\delta(x)/\delta(y) = h(x, y)$ for all $x, y \in (0, 1)$ such that $x + y < 1$. To see this, notice that:

1. If $x, y \in (0, 3/4]$, then from (5) it follows that

$$\frac{\delta(x)}{\delta(y)} = \frac{h(x, 1/5)}{h(y, 1/5)} = h(x, y).$$

2. If $x \in (3/4, 1)$ (and thus $y \in (0, 1/4)$), then

$$\frac{\delta(x)}{\delta(y)} = \frac{\frac{h(x, (1-x)/5)}{h(1/5, (1-x)/5)}}{h(y, 1/5)} = \frac{h(x, (1-x)/5)}{h(1/5, (1-x)/5)h(y, 1/5)},$$

where $h(1/5, (1-x)/5) = 1/h((1-x)/5, 1/5)$ and, given (5), $h(x, (1-x)/5) = h(x, y)/h((1-x)/5, y)$, so

$$\frac{\delta(x)}{\delta(y)} = \frac{h(x, y)}{h((1-x)/5, y)} \frac{h((1-x)/5, 1/5)}{h(y, 1/5)} = \frac{h(x, y)}{h((1-x)/5, y)} h((1-x)/5, y) = h(x, y),$$

where the second equality also follows from (5).

3. If $y \in (3/4, 1)$ (and thus $x \in (0, 1/4)$), then $\delta(x)/\delta(y) = (\delta(y)/\delta(x))^{-1} = (h(y, x))^{-1} = h(x, y)$, where the second equality follows from the previous case.

We have then that for every s_{*f} and distinct $i, j \in N_f(s_{*f})$ such that $s_{if} + s_{jf} < 1$, $\gamma_{if}(s_{*f})/\gamma_{jf}(s_{*f}) = \delta(s_{if})/\delta(s_{jf})$. This equality also automatically holds when $j \in N_f(s_{*f})$ but $i \notin N_f(s_{*f})$. Therefore, for every s_{*f} such that $|N_f(s_{*f})| \geq 3$ and every $j \in N_f(s_{*f})$

$$1 = \sum_{i \in N} \gamma_{if}(s_{*f}) = \sum_{i \in N} \frac{\delta(s_{if})}{\delta(s_{jf})} \gamma_{jf}(s_{*f}) = \frac{\gamma_{jf}(s_{*f})}{\delta(s_{jf})} \sum_{i \in N} \delta(s_{if}) \implies \gamma_{jf}(s_{*f}) = \frac{\delta(s_{jf})}{\sum_{i \in N} \delta(s_{if})}.$$

Also, for $j \notin N_f(s_{*f})$, it automatically holds that $\gamma_{jf}(s_{*f}) = \delta(s_{jf})/\sum_{i \in N} \delta(s_{if}) = \delta(0)/\sum_{i \in N} \delta(s_{if}) = 0$. **Q.E.D.**

Proof of Lemma 2. Take any WAPP and rank-preserving strategic plan and let γ_{*f} be its control power function. With shareholders ordered by descending firm f stake, rank preservation gives $\gamma_{1f}(s_{*f}) \geq \dots \geq \gamma_{n_f f}(s_{*f}) \geq 0$. Also, $s_{if} = 0$ and thus $\gamma_{if}(s_{*f}) = 0$ for every $i > n_f$. For every $k < n_f$, let $\xi_k := \gamma_{kf}(s_{*f}) - \gamma_{k+1, f}(s_{*f})$ and $\xi_{n_f} := \gamma_{n_f f}(s_{*f}) \geq 0$, so that $\sum_{k=i}^{n_f} \xi_k = \gamma_{if}(s_{*f})$ for every $i \leq n_f$. If $k \leq n_f$ and $k \notin K$, then $s_{kf} = s_{k+1, f}$, so by rank preservation, $\xi_k = 0$. Therefore, $\gamma_{if}(s_{*f}) = \sum_{k \in K: k \geq i} \xi_k$, so

$$\lambda_{fg}(s) = \frac{\sum_{i=1}^{n_f} \gamma_{if}(s_{*f}) s_{ig}}{\sum_{i=1}^{n_f} \gamma_{if}(s_{*f}) s_{if}} = \frac{\sum_{i=1}^{n_f} s_{ig} \sum_{k \in K: k \geq i} \xi_k}{\sum_{i=1}^{n_f} s_{if} \sum_{k \in K: k \geq i} \xi_k} = \frac{\sum_{k \in K} \xi_k \sum_{i=1}^k s_{ig}}{\sum_{k \in K} \xi_k \sum_{i=1}^k s_{if}} = \frac{\sum_{k \in K} \tilde{\xi}_k \Lambda_k}{\sum_{k \in K} \tilde{\xi}_k},$$

where $\tilde{\xi}_k := \xi_k \sum_{i=1}^k s_{if} \geq 0$. Thus, $\lambda_{fg}(s) \in [\underline{\lambda}, \bar{\lambda}]$ as a weighted average of $(\Lambda_k)_{k \in K}$.

Conversely, for every $k \in K$, let γ_{*f}^k be the control power function with $\gamma_{if}^k(s_{*f}) := \mathbb{I}(i \leq k)/k$. The resulting WAPP strategic plan is rank-preserving. Also, under control

power function $\gamma_{*f}^{\underline{\kappa}}$, $\lambda_{fg}(s) = \Lambda_{\underline{\kappa}}$. Particularly, under control power function $\gamma_{*f}^{\underline{\kappa}}$ with $\underline{\kappa} \in \arg \min_{k \in K} \Lambda_k$, $\lambda_{fg}(s) = \underline{\lambda}$. Similarly, under control power function $\gamma_{*f}^{\bar{\kappa}}$ with $\bar{\kappa} \in \arg \max_{k \in K} \Lambda_k$, $\lambda_{fg}(s) = \bar{\lambda}$. By continuity, intermediate $\lambda_{fg}(s)$'s are attained by convex combinations of $\gamma_{*f}^{\underline{\kappa}}$ and $\gamma_{*f}^{\bar{\kappa}}$, which are also rank-preserving. **Q.E.D.**

Proof of Proposition 6. Since $q_f^t > 0$ for every t , $q_f^t = (a - c_f^*)/(2b) - (1 + \lambda_{fg}^m)q_g^t/2$. Let $\bar{q}_g := \sum_{t=1}^T q_g^t/T$. Substituting for q_f^t , the residual of the model with weight x on firm g 's profit and marginal cost c_f is $(c_f - c_f^*)/(2b) - (\lambda_{fg}^m - x)q_g^t/2$, so

$$\inf_{c_f} \sum_{t=1}^T \left(\frac{c_f - c_f^*}{2b} - \frac{(\lambda_{fg}^m - x)q_g^t}{2} \right)^2 = \frac{(\lambda_{fg}^m - x)^2}{4} \sum_{t=1}^T (q_g^t - \bar{q}_g)^2,$$

the sum of squares being minimized at $c_f = c_f^* + b(\lambda_{fg}^m - x)\bar{q}_g$. Setting $x = 0$ gives Σ^{OPM} . By Lemma 2, taking the infimum over $x \in [\underline{\lambda}, \bar{\lambda}]$ gives Σ^{WAPP} . The sample has $q_f^t \neq q_f^{t'}$ for some t, t' , so $q_g^t \neq q_g^{t'}$ for those two observations. Thus, $\sum_{t=1}^T (q_g^t - \bar{q}_g)^2 > 0$, and $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$ if and only if $(\lambda_{fg}^m)^2 < \inf_{x \in [\underline{\lambda}, \bar{\lambda}]} (\lambda_{fg}^m - x)^2$, and the result follows. **Q.E.D.**

Proof of Proposition 7. By Lemma 2, $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$ at both firms requires $\lambda^m < 1/2$. To see that any such λ^m can be attained, consider the following ownership structure. Take arbitrary $\lambda \in (0, 1/2)$ and $L \in \mathbb{N}$. The ownership structure s is symmetric with $2L + 1$ pairwise disjoint groups of shareholders owning firms f and g : $N = D \cup \bigcup_{\ell=0}^{L-1} (N_f^\ell \cup N_g^\ell)$ with $|D| = 1$ and $|N_f^\ell| = |N_g^\ell| = \nu_\ell$. Group D is comprised of a unique large diversified shareholder with large stakes in both firms: $s_{if} = s_{ig} = \bar{s}$ for $i \in D$. $s_{if} = \lambda^\ell \underline{s}$ and $s_{ig} = \lambda^{\ell+1} \underline{s}$ for every $i \in N_f^\ell$ with $\bar{s} > \underline{s} > 0$. Similarly, $s_{ig} = \lambda^\ell \underline{s}$ and $s_{if} = \lambda^{\ell+1} \underline{s}$ for every $i \in N_g^\ell$. $A_\ell := \sum_{j=0}^\ell \nu_j \lambda^j \underline{s}$ is the number of firm f shares held by groups N_f^0 through N_f^ℓ . Each firm's shares sum up to one if and only if $\bar{s} + (1 + \lambda)A_{L-1} = 1$. Also, $\lambda_{i;fg} = \lambda$ for every $i \in \bigcup_\ell N_f^\ell$, so if $A_{L-1} \geq 1/2$, then $\lambda_{fg}^m = \lambda_{gf}^m = \lambda$.

Every shareholder holds a positive stake in firm f , so $n_f = n = 1 + 2 \sum_{j=0}^{L-1} \nu_j$. In descending order, firm f shareholders' shares take the following n values: $\bar{s} > \underline{s} = \dots = \underline{s} > \lambda \underline{s} = \dots = \lambda \underline{s} > \dots > \lambda^L \underline{s} = \dots = \lambda^L \underline{s}$, where for every $i \in N_f^\ell \cup N_g^{\ell-1}$, $s_{if} = \lambda^\ell \underline{s}$ (with $N_g^{-1} := \emptyset$) and for every $i \in N_g^{L-1}$, $s_{if} = \lambda^L \underline{s}$. For every $\ell \in \{0, \dots, L-1\}$, let $k_\ell := 1 + \sum_{j=0}^\ell \nu_j + \sum_{j=0}^{\ell-1} \nu_j$ with $k_{-1} = 1$ and $k_{L-1} = n - \nu_{L-1}$, so that shareholders $k_{\ell-1} + 1, \dots, k_\ell$ are the members of $N_f^\ell \cup N_g^{\ell-1}$, and shareholders $k_{L-1} + 1, \dots, n$ are the members of N_g^{L-1} . Then, $K = \{1, k_0, \dots, k_{L-1}, n\}$, $\Lambda_1 = \Lambda_n = 1$, and $\Lambda_{k_\ell} = (\bar{s} + \lambda A_\ell + A_{\ell-1})/(\bar{s} + A_\ell + \lambda A_{\ell-1}) < 1$ for every ℓ since $A_\ell > A_{\ell-1}$ (with $A_{-1} := 0$). By Proposition 6, $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$ if and only if $\underline{\lambda} > 2\lambda$, or equivalently, $\bar{s}(1 - 2\lambda) + (1 - 2\lambda^2)A_{\ell-1} > \lambda A_\ell$ for every $\ell \in \{0, \dots, L-1\}$. Substituting $A_\ell = A_{\ell-1} + m_\ell$, where $m_\ell := A_\ell - A_{\ell-1} = \nu_\ell \lambda^\ell \underline{s}$ is the number of firm f shares held by group N_f^ℓ , these L inequalities read $\lambda m_\ell < (1 - 2\lambda)[\bar{s} + (1 + \lambda)A_{\ell-1}]$. It remains to choose L , $\underline{s}$ and $(\nu_\ell)_{\ell=0}^{L-1} \in \mathbb{N}^L$ with $\bar{s} = 1 - (1 + \lambda)A_{L-1} > \underline{s} > 0$ and

$A_{L-1} \geq 1/2$ satisfying these L inequalities.

Define $r := (1 - 2\lambda^2)/\lambda > 1$, pick $a, b \in \mathbb{N}$ such that $a/b \in (1/\lambda, r/\lambda)$, and define $t := \lambda a/b \in (1, r)$. Since $t > 1$, we can pick L large enough such that $(t^L - 1)/(t - 1) > \lambda/[(1 - 2\lambda)(1 - \lambda)]$. Define $M := (t - 1)/[2(t^L - 1)]$ and set $\underline{s} := M/(2b^{L-1})$ and $\nu_\ell := 2b^{L-1-\ell}a^\ell \in \mathbb{N}$ for every ℓ . Then, $m_\ell \equiv \nu_\ell \lambda^\ell \underline{s} = Mt^\ell$, and therefore $A_\ell = M \sum_{j=0}^\ell t^j = M(t^{\ell+1} - 1)/(t - 1)$, $A_{L-1} = 1/2$, and $\bar{s} = (1 - \lambda)/2$. Also, $\underline{s} \leq M/2 \leq 1/4 < (1 - \lambda)/2 = \bar{s}$. Finally, for every ℓ ,

$$\lambda m_\ell - (1 - 2\lambda)(1 + \lambda)A_{\ell-1} = \lambda M \left[t^\ell - \frac{(r - 1)(t^\ell - 1)}{t - 1} \right] \leq \lambda M < (1 - 2\lambda)\bar{s},$$

where we have used $\lambda(r - 1) = (1 - 2\lambda)(1 + \lambda)$, $r > t > 1$, and $(t^L - 1)/(t - 1) > \lambda/[(1 - 2\lambda)(1 - \lambda)]$, so all L inequalities are satisfied.

Last, $\lambda^m = 0$ is attained for $n = 5$, $s_{1f} = s_{1g} = 1/3$, $s_{if} = 1/3$, $s_{ig} = 0$ for $i \in \{2, 3\}$, and $s_{if} = 0$ and $s_{ig} = 1/3$ for $i \in \{4, 5\}$. Then, $n_f = 3$, $K = \{3\}$, and $\underline{\lambda} = \Lambda_3 = 1/3$, while $\lambda_{1;fg} = 1$ and $\lambda_{2;fg} = \lambda_{3;fg} = 0$ give $\lambda_{fg}^m = 0$. Proposition 6 therefore gives $\Sigma^{\text{OPM}} < \Sigma^{\text{WAPP}}$ for firm f and, by symmetry, for firm g as well.

Over $\lambda^m \in [0, 1/2)$, $\mu_{\text{increase}} = 2 - 6/(3 + \lambda^m)$, $\text{CS}_{\text{loss}} = 1 - 9/(3 + \lambda^m)^2$, and $\text{TS}_{\text{loss}} = 1 - 9(2 + \lambda^m)/[2(3 + \lambda^m)^2]$ are continuous and increasing in λ^m , so their ranges as λ^m varies between 0 and 1/2 are $[0, 2/7)$, $[0, 13/49)$, and $[0, 4/49)$. **Q.E.D.**

Online Appendix

Corporate control under common ownership

Orestis Vravosinos

B The weighted average profit weight strategic plan (WAPW)

In Brito et al.'s (2023) voting model, when the profit relevance of shareholder bias parameter is equal to 1, the authors frame the strategic plan as—what I call—a weighted average profit weight strategic plan (WAPW).

Definition 15. Firm f 's strategic plan R_f is a weighted average profit weight (WAPW) if there exists a control power function $\hat{\gamma}_{*f} : \Delta^n \rightarrow \Delta^n$ such that for every $s \in S$, $\alpha_{-f} \in \times_{g \neq f} \Delta(A_g)$, and $\pi \in \Pi$,

- (i) the firm's Edgeworth coefficient of effective sympathy towards another firm is equal to the weighted average of its shareholders' Edgeworth coefficients:

$$R_f(\alpha_{-f}, s; \pi) = \arg \max_{\alpha_f \in \Delta(A_f)} \left\{ \pi_f(\alpha_f, \alpha_{-f}) + \sum_{g \in M \setminus \{f\}} \left(\sum_{i \in N_f(\gamma_{*f})} \hat{\gamma}_{if}(s_{*f}) \lambda_{i;fg} \right) \pi_g(\alpha_f, \alpha_{-f}) \right\},$$

where $N_f(\gamma_{*f}(s_{*f})) \equiv \{i \in N : \hat{\gamma}_{if}(s_{*f}) > 0\}$,

- (ii) control exclusive to shareholders: For every $i \in N$, $s_{if} = 0 \implies \hat{\gamma}_{if}(s_{*f}) = 0$.

In WAPW, the weight that the manager of firm f places on firm g 's profit is a weighted average of the weights $\{\lambda_{i;fg}\}_{i \in N_f(s_{*f})}$ that the shareholders of firm f would want firm f to use. This still is WAPP, since it can be written as

$$R_f(\alpha_{-f}, s) = \arg \max_{\alpha_f \in \Delta(A_f)} \left\{ \sum_{i \in N_f(\hat{\gamma}_{*f})} \gamma_{if}(s_{*f}) u_i(\alpha_f, \alpha_{-f}, s_{i*}) \right\},$$

where for every shareholder i of firm f

$$\gamma_{if}(s_{*f}) := \frac{\hat{\gamma}_{if}(s_{*f})/s_{if}}{\sum_{i \in N_f(\hat{\gamma}_{*f})} \hat{\gamma}_{if}(s_{*f})/s_{if}}.$$

Thus, a strategic plan is WAPP if and only if it is WAPW. The novelty is that the WAPW parametrizations considered in Brito et al. (2023) give rise to γ 's that are not standard in the literature. If all shares have voting rights, proportional $\hat{\gamma}$'s give rise to

$$\gamma_{if}(s_{*f}) = \begin{cases} 1/|N_f(s_{*f})| & \text{if } s_{if} > 0 \\ 0 & \text{if } s_{if} = 0. \end{cases}$$

That is, firm f maximizes the *unweighted* average of its shareholders' portfolio profits. Section 4.5 discusses the implications of this WAPP formulation. Banzhaf $\hat{\gamma}$'s give rise to

$$\gamma_{if}(s_{*f}) := \begin{cases} \frac{\gamma_{if}^B(s_{*f})/s_{if}}{\sum_{j \in N_f(s_{*f})} \gamma_{jf}^B(s_{*f})/s_{jf}} & \text{if } s_{if} > 0 \\ 0 & \text{if } s_{if} = 0. \end{cases}$$

where

$$\gamma_{if}^B(s_{*f}) = \frac{\left| \left\{ T \in 2^N : \sum_{k \in T} s_{kf} \geq 1/2 > \sum_{k \in T \setminus \{i\}} s_{kf} \right\} \right|}{\sum_{t \in N} \left| \left\{ T \in 2^N : \sum_{k \in T} s_{kf} \geq 1/2 > \sum_{k \in T \setminus \{t\}} s_{kf} \right\} \right|}.$$

C The random dictatorship disagreement payoff function

The random dictatorship specification of the disagreement payoff function posits that in case of disagreement, there is random dictatorship: With some exogenous probability, each shareholder of the firm is chosen to implement her most preferred strategy.

Definition 16. The disagreement payoff function d_{*f} is a random dictatorship (RD) disagreement payoff function if there exist a lottery weight function $\mu_{*f} : \Delta^n \rightarrow \Delta^n$ and a choice function in case of disagreement $\alpha_f^d : \times_{g \neq f} \Delta(A_g) \times \{v \in \mathbb{R}_+^m : v_f = 1\} \rightarrow \Delta(A_f)$ such that for every $s \in S$ and $\alpha_{-f} \in \times_{g \neq f} \Delta(A_g)$,

- (i) the strategy $\alpha_f^d(\alpha_{-f}, v)$ maximizes the payoff of a shareholder with relative holdings v in the firms: For every $v \in \{v' \in \mathbb{R}_+^m : v'_f = 1\}$,

$$\alpha_f^d(\alpha_{-f}, v) \in \arg \max_{\alpha_f \in \Delta(A_f)} \sum_{g \in M} v_g \pi_g(\alpha_f, \alpha_{-f}),$$

- (ii) disagreement payoffs are derived from random dictatorship:

$$d_{*f}(\alpha_{-f}, s) = \sum_{i \in N} \mu_{if}(s_{*f}) u \left(\alpha_f^d(\alpha_{-f}, \lambda_{i;f*}), \alpha_{-f}, s \right),$$

- (iii) control is exclusive to shareholders: For every $i \in N$, $s_{if} = 0 \implies \mu_{if}(s_{*f}) = 0$.

D Equilibrium existence

Under NB, the equilibrium is a Nash equilibrium in Nash bargains. Particularly, the oligopoly game can be seen as a generalized game where each firm's strategy set depends on the other firms' strategies. Namely, when the other firms play α_{-f} , firm f can choose a strategy in $B_f(\alpha_{-f}, s; \pi)$, because it needs to make sure that each controlling shareholder

achieves at least her disagreement payoff. Proposition 8 provides sufficient conditions for existence of a pure equilibrium of this generalized game.

Proposition 8. Fix some $\pi \in \Pi$ and $s \in S$. Assume that for every firm $f \in M$,

- (i) A_f is a nonempty, compact and convex subset of a Euclidean space,
- (ii) $\pi_f(a)$ is continuous in a ,
- (iii) for each $i \in N$, $d_{if}(a_{-f}, s)$ is continuous in a_{-f} ,
- (iv) $B_f^P(a_{-f}, s)$ is lower hemicontinuous in a_{-f} over $a_{-f} \in A_{-f}$,
- (v) $u_i(a_f, a_{-f}, s_{i*})$ is concave in a_f over A_f for every $i \in N_f(\beta_{*f}(s_{*f}))$ and every $a_{-f} \in A_{-f}$,

where $B_f^P(a_{-f}, s) := \{a_f \in A_f : u_i(a_f, a_{-f}, s_{i*}) \geq d_{if}(a_{-f}, s) \ \forall i \in N_f(\beta_{*f}(s_{*f}))\}$. Then, a pure Nash equilibrium in Nash bargains exists.

Assumption (v) guarantees that the Nash product is quasiconcave in a_f . For sufficient conditions for assumption (iv) of Proposition 8 to hold, see Proposition 4.2 in Dutang (2013), which is an application of Theorem 5.9 in Rockafellar and Wets (1997), Proposition 4.3 in Dutang (2013), Theorem 13 in Hogan (1973), Corollary 2 in Maćkowiak (2006), and Claim 2 in Banks and Duggan (2004).

E Noisy common ownership hypothesis testing

This section describes how the analysis of section 5 applies to a noisy setting. The only difference from section 5 is that firm f 's marginal cost in observation t is $c_f^t := c_f^* + \varepsilon_f^t$, where $c_f^* > 0$ and $(\varepsilon_f^t)_{t=1}^T$ are mean-zero idiosyncratic cost shocks. Therefore, $q_f^t = R_f(q_g^t, \varepsilon_f^t, s) \equiv \max\{(a - c_f^* - \varepsilon_f^t)/(2b) - (1 + \lambda_{fg}^m)q_g^t/2, 0\}$. $\Sigma^{\text{OPM}} := \inf_{c_f} \sum_{t=1}^T (q_f^t - R_f^{\text{OPM}}(q_g^t, s; c_f))^2$ and $\Sigma^{\text{WAPP}} := \inf_{c_f, \gamma_{*f}} \sum_{t=1}^T (q_f^t - R_f^{\text{WAPP}}(q_g^t, s; c_f))^2$. Lemma 2 still applies. If $(q_g^t, \varepsilon_f^t)_{t=1}^T$ are i.i.d. with $0 < \text{Var}(q_g^t) < \infty$ and $\text{Var}(\varepsilon_f^t) < \infty$, $(q_g^t)_{t=1}^T$ and $(\varepsilon_f^t)_{t=1}^T$ are independent, and $q_f^t > 0$ for every t , we have that

$$\begin{aligned} \Sigma^{\text{OPM}} &= \inf_{c_f} \sum_{t=1}^T \left(\frac{c_f - c_f^* - \varepsilon_f^t}{2b} - \frac{\lambda_{fg}^m}{2} q_g^t \right)^2 = \sum_{t=1}^T \left(\frac{\bar{\varepsilon}_f - \varepsilon_f^t}{2b} - \frac{\lambda_{fg}^m}{2} (q_g^t - \bar{q}_g) \right)^2 \\ &= \frac{\sum_{t=1}^T (\varepsilon_f^t - \bar{\varepsilon}_f)^2}{4b^2} + \frac{\lambda_{fg}^m}{2b} \sum_{t=1}^T (\varepsilon_f^t - \bar{\varepsilon}_f)(q_g^t - \bar{q}_g) + \frac{(\lambda_{fg}^m)^2}{4} \sum_{t=1}^T (q_g^t - \bar{q}_g)^2, \quad \text{and} \\ \Sigma^{\text{WAPP}} &= \inf_{c_f} \sum_{t=1}^T \left(\frac{c_f - c_f^* - \varepsilon_f^t}{2b} - \frac{\lambda_{fg}^m - x}{2} q_g^t \right)^2 \\ &= \inf_{x \in [\underline{\lambda}, \bar{\lambda}]} \sum_{t=1}^T \left(\frac{\bar{\varepsilon}_f - \varepsilon_f^t}{2b} - \frac{\lambda_{fg}^m - x}{2} (q_g^t - \bar{q}_g) \right)^2 \end{aligned}$$

$$= \inf_{x \in [\underline{\lambda}, \bar{\lambda}]} \left\{ \frac{\sum_{t=1}^T (\varepsilon_f^t - \bar{\varepsilon}_f)^2}{4b^2} + \frac{\lambda_{fg}^m - x}{2b} \sum_{t=1}^T (\varepsilon_f^t - \bar{\varepsilon}_f)(q_g^t - \bar{q}_g) + \frac{(\lambda_{fg}^m - x)^2}{4} \sum_{t=1}^T (q_g^t - \bar{q}_g)^2 \right\},$$

where $\bar{q}_g := \sum_{t=1}^T q_g^t / T$ and $\bar{\varepsilon}_f := \sum_{t=1}^T \varepsilon_f^t / T$. Therefore,

$$\begin{aligned} \frac{\Sigma^{\text{WAPP}} - \Sigma^{\text{OPM}}}{T} &= \inf_{x \in [\underline{\lambda}, \bar{\lambda}]} \left\{ -\frac{x}{2b} \frac{\sum_{t=1}^T (\varepsilon_f^t - \bar{\varepsilon}_f)(q_g^t - \bar{q}_g)}{T} + \frac{x(x - 2\lambda_{fg}^m)}{4} \frac{\sum_{t=1}^T (q_g^t - \bar{q}_g)^2}{T} \right\} \\ &\xrightarrow{p} \frac{\underline{\lambda}(\underline{\lambda} - 2\lambda_{fg}^m)}{4} \text{Var}(q_g^t) > 0 \end{aligned}$$

by the law of large numbers, given that ε_f^t and q_g^t are independent and $\underline{\lambda} > 2\lambda_{fg}^m$.

F Competitive effects of common ownership and policy implications

This section shows that WAPP and NB can give rise to significantly different theoretical predictions and policy implications. Specifically, I look at how market outcomes change as a shareholder varies the degree of diversification of a fixed number of shares across the industry.

Consider a homogeneous product Cournot duopoly ($m = 2$) with 3 shareholders ($n = 3$), linear inverse demand $P(Q) = \max\{10 - Q, 0\}$ and symmetric linear cost functions $C_1(q_1) = q_1$, $C_2(q_2) = q_2$. We will consider two cases: (i) both firms' strategic plans are WAPP with proportional control and (ii) both firms' strategic plans are NB with $\beta_{*f}(s_{*f}) = s_{*f}$ and random dictatorship disagreement payoffs with $\mu_{*f}(s_{*f}) = s_{*f}$. Let the ownership matrix be

$$s = \begin{bmatrix} s_{11} & 0.45 - s_{11} \\ 1 - s_{11} & 0 \\ 0 & 0.55 + s_{11} \end{bmatrix},$$

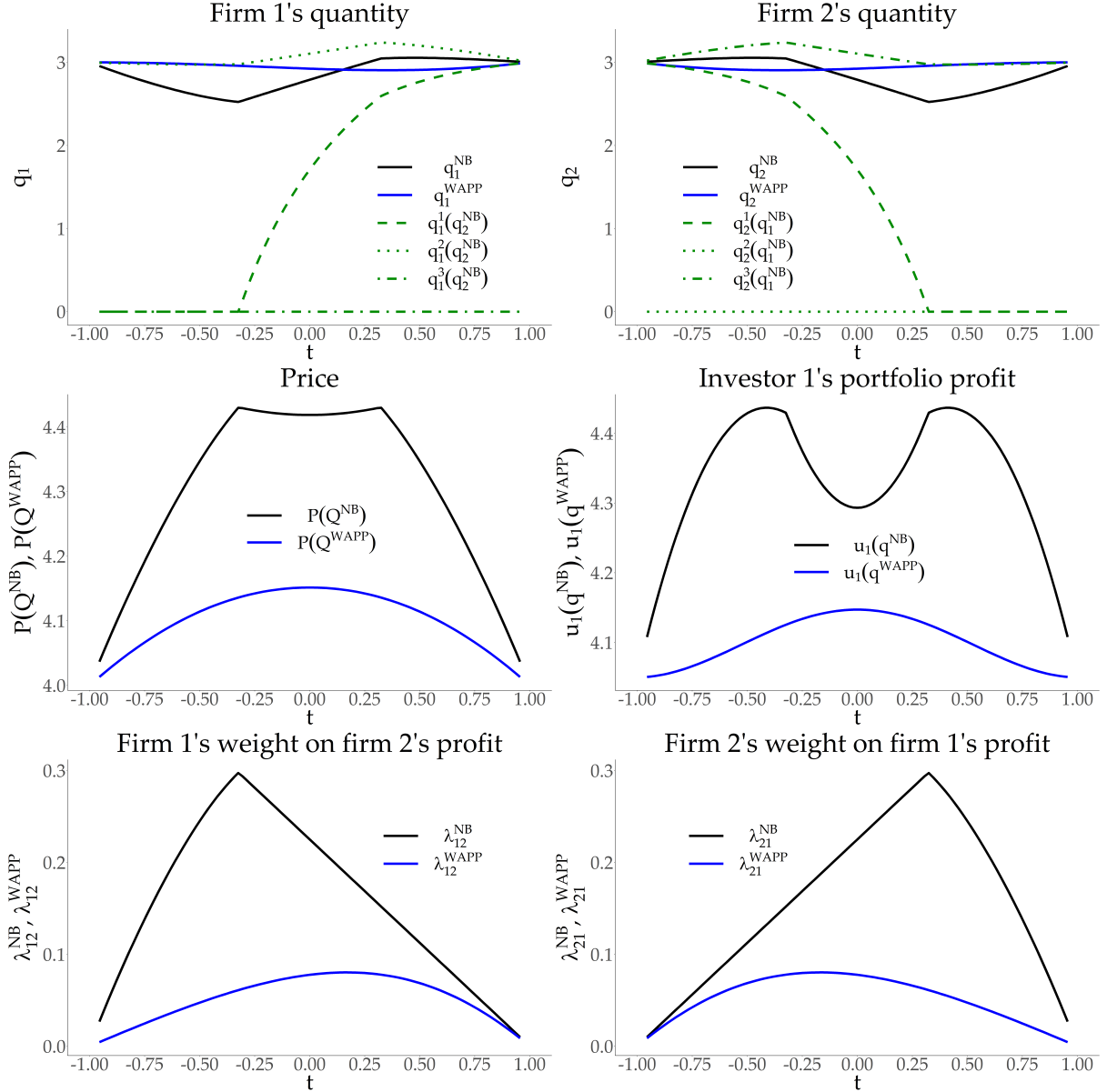
which is indexed by the shares s_{11} of shareholder 1 in firm 1.

The two firms are equally efficient and shareholder 1 (e.g., a large fund) can choose how to distribute her total holdings of 0.45 in the industry between the two firms. Shareholders 2 and 3 are passive in that they are indifferent towards the capital they invest in the firms. The fund can buy shares of either firm at the same price and the rest of the capital is provided by shareholders 2 and 3. Define the normalized value $t := (s_{11} - 0.225) / 0.225 \in [-1, 1]$ measuring what firm the fund's holdings are concentrated in. The closer t is to 0, the higher is the fund's diversification; for $t = 0$ the equilibrium is symmetric. As t increases shareholder 1's holdings become more concentrated in firm 1.

Think of a policy that limits the degree of common ownership a shareholder can have within the industry; it specifies some $\tau \in [0, 1]$ and requires that $t \in [-1, -\tau] \cup [\tau, 1]$.

Figure 2 shows equilibrium results under NB and WAPP.

Figure 2: Equilibrium with a large fund and two undiversified passive shareholders for varying levels of diversification by the fund



Note: black lines represent equilibrium values under NB; blue ones under WAPP. Green lines show the most preferred quantity of each shareholder for each firm with the competitor's quantity taken as given (fixed at its equilibrium value). The bottom two panels plot $\lambda_{12}, \lambda_{21}$ (under WAPP) and $\tilde{\lambda}_{12}, \tilde{\lambda}_{21}$ (under NB).

If the fund only cares to maximize its portfolio profit, then under WAPP it will choose t as close to 0 as possible. Thus, the price is decreasing in the restrictiveness τ of the policy. However, under NB the fund picks t as close as possible to either of the two peaks (in its portfolio profit) as possible, so that the price is first constant and then decreasing in τ . Therefore, a policy that is effective in increasing consumer welfare under WAPP

may be ineffective under NB.²⁹

Consider now an alternate scenario where the fund only cares to maximize its portfolio diversification, that is $\min |t|$, in order for example to mitigate risk or track an industry index. Then, under WAPP, the price is decreasing in τ . However, under NB, the price is first increasing and then decreasing in τ . Thus, a policy that is effective under WAPP may in fact harm consumer welfare under NB.

The differences in predictions between WAPP and NB are due to the differences (between the two strategic plans) in magnitudes of the various channels through which a change in t affects equilibrium outcomes. As t changes, both the fund's preferences and the division of power within each firm change.

Under WAPP, as t (*i.e.*, s_{11}) increases, the degree to which the fund wants firm 1 (resp. 2) to internalize firm 2's (resp. 1's) profits decreases (resp. increases), which tends to shift production towards firm 1. On the other hand, as t increases shareholder 2's control of firm 1 decreases, and shareholder 3's control of firm 2 increases, which tend to shift production towards firm 2. Under WAPP, around $t = 0$, the latter effects dominate. Consequently, firm 2's quantity increases with t , while the quantity of firm 1 decreases, making it unprofitable for the fund to pick $t \neq 0$. Also, firm 2's quantity increases faster than firm 1's quantity decreases with t (around $t = 0$), and the price has a global maximum at $t = 0$ under WAPP.

However, under NB, as t increases (around $t = 0$), production shifts towards firm 1, which is in the interest of the fund when $t > 0$. This makes it profitable for the fund to pick $t \neq 0$. Also, firm 1's quantity increases more slowly than firm 2's quantity decreases with t (around $t = 0$), so that the price has a local minimum at $t = 0$ under NB.

Similarly, based on WAPP a consumer-welfare-maximizing regulator would want to block a trade that brings t from -0.25 to 0 , even though this trade would increase consumer welfare under NB.

Last, notice that the graphs of control weights γ and $\tilde{\gamma}$ differ between WAPP and NB. These weights capture the extent to which changes in shareholder preferences (e.g., due to a stock trade) will be accommodated by each firm. Thus, the WAPP and NB models will give different predictions regarding stock trade effects.

G Application: homogeneous product Cournot oligopoly

This section characterizes the Nash-in-Nash equilibrium of a homogeneous product Cournot oligopoly and studies how changes in corporate control affect equilibrium outcomes. Given that the FOCs under NB are analogous to those under WAPP (see section 3.2), the analysis is also valid under WAPP.

²⁹Remember that consumer surplus is increasing in the total quantity (and thus decreasing in the price) in a homogeneous product market.

G.1 A Nash-in-Nash model of Cournot oligopoly with common ownership

The firms produce a homogeneous good. Each firm f chooses its production quantity q_f simultaneously with the other firms. Denote by $w_f \equiv q_f/Q$ firm f 's market share of the total quantity $Q := \sum_{g \in M} q_g$. q_{-f} denotes the production profile of the firms other than f , and $Q_{-f} := \sum_{g \neq f} q_g$. Firm f 's production cost is given by the twice-differentiable function $C_f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ with $C'_f(q_f) > 0$ globally.

The twice-differentiable inverse demand function $P(Q)$ satisfies $P'(Q) < 0 \forall Q \in [0, \bar{Q}]$, where $\bar{Q} \in (0, +\infty]$ is such that $P(Q) > 0 \iff Q \in [0, \bar{Q}]$. $\eta(Q) := -P/(QP')$ denotes the elasticity of demand. $E(Q) := -P''(Q)Q/P'(Q)$ denotes the absolute value of the elasticity of the slope of inverse demand. Firm f 's profit is given by $\pi_f(q) := q_f P(Q) - C_f(q_f)$.

Define the following index of the weight firm f places on other firms' profits

$$\bar{\lambda}_f(q, s) := \sum_{g \in M \setminus \{f\}} w_g \tilde{\lambda}_{fg}(q_{-f}, s) \equiv \sum_{g \in M \setminus \{f\}} w_g \frac{\sum_{i \in N_f(\beta_{*f}(s_{*f}))} \tilde{\gamma}_{if}(q_{-f}, s) s_{ig}}{\sum_{i \in N_f(\beta_{*f})} \tilde{\gamma}_{if}(q_{-f}, s) s_{if}}.$$

Similarly, for each firm f and each shareholder i of firm f define $\bar{\lambda}_{i,f}(q, s_{i*}) := \sum_{g \in M \setminus \{f\}} w_g \lambda_{i,fg}$, an index of the weight shareholder i wants firm f to place "on average" on other firms' profits.

Define also the bargaining-adjusted (i) Herfindahl-Hirschman Index (HHI) of market shares, (ii) MHHI Δ , and (iii) modified HHI, (iv) weighted average Lerner index LI, respectively given by

$$\begin{aligned} \text{HHI}(q) &:= \sum_{g \in M} w_g^2, \quad \text{MHHI}\Delta(q, s) := \sum_{g \in M} w_g \bar{\lambda}_g(q, s), \\ \text{MHHI}(q, s) &:= \text{HHI}(q) + \text{MHHI}\Delta(q, s), \quad \bar{\text{LI}}(q) := \sum_{g=1}^m w_g \frac{P(Q) - C'_g(q_g)}{P(Q)}. \end{aligned}$$

G.2 The firm's problem in a homogeneous product Cournot market

Lemma 3 provides conditions under which a shareholder's portfolio profit is strictly concave in a firm's quantity.

Lemma 3. Fix a firm $f \in M$ and a shareholder $i \in N_f(s_{*f})$. If for every quantity profile q such that $Q < \bar{Q}$, $E(Q)(\bar{\lambda}_{i,f}(q, s_{i*}) + w_f) < 2 - C''_f(q_f)/P'(Q)$, then for any q_{-f} , $u_i(q, s_{i*})$ is strictly concave in q_f for every q_f such that $Q < \bar{Q}$.

A sufficient condition is $\max\{E(Q) \max_{g \in M} \lambda_{i,fg}, 0\} < 2 - C''_f(q_f)/P'(Q)$ for every q_f and Q such that $q_f \leq Q < \bar{Q}$. A simpler sufficient condition is $C''_f \geq 0$ and $E(Q) < 2/\max_{g \in M} \lambda_{i,fg}$. These conditions deliver the strict concavity assumed in Lemma 4 at every quantity profile with $Q < \bar{Q}$. Lemma 4 characterizes the firm's problem.

Lemma 4. Assume that there exists $\bar{q} > 0$ such that $P(q) < C_f(q)/q$ for every $q > \bar{q}$ and every firm $f \in M$. Fix a firm $f \in M$ and q_{-f} and let the strategic plan R_f be NB. Assume that for every shareholder $i \in N$, $u_i(q, s_{i*})$ is strictly concave in q_f . Then, the following statements are true:

- (i) $B_f^P(q_{-f}, s) \equiv \{q_f \in A_f : u_i(q_f, q_{-f}, s_{i*}) \geq d_{if}(q_{-f}, s) \text{ for every } i \in N_f(\beta_{*f}(s_{*f}))\}$ is a closed interval,
- (ii) $R_f(q_{-f}, s)$ is a singleton,
- (iii) over $B_f^P(q_{-f}, s)$, the Nash product is increasing (resp. decreasing) in q_f for q_f $\overset{(\text{resp.})}{<}$ $R_f(q_{-f}, s)$.

G.3 Nash-in-Nash equilibrium characterization

Let $\tilde{S} \subseteq S$ be an open subset of S such that for every $s \in \tilde{S}$, there is a unique and interior equilibrium q^* where $u(\text{NB}_{\beta_{*f}, d_{*f}}(q_{-f}^*, s), q_{-f}^*, s) \gg d_{*f}(q_{-f}^*, s)$ for every firm $j \in M$. $q^* : \tilde{S} \rightarrow \mathbb{R}_{++}^m$ returns this equilibrium as a function of s . We will sometimes simply write q^* instead of $q^*(s)$. Similarly, write $Q^* \equiv \sum_{g \in M} q_g^*$, $w_f^* := q_f^*/Q^*$. To simplify notation, define also $\gamma_{if}^*(s) := \tilde{\gamma}_{if}(q_{-f}^*(s), s)$, $\lambda_{fg}^*(s) := \tilde{\lambda}_{fg}(q_{-f}^*(s), s)$, $\bar{\lambda}_f^*(s) := \bar{\lambda}_f(q^*(s), s)$, $\bar{\lambda}_{i,f}^*(s) := \bar{\lambda}_{i,f}(q^*(s), s_{i*})$ for every shareholder $i \in N$ and pair of distinct firms $f, g \in M$. These functions give the equilibrium values of the corresponding objects as functions of the ownership matrix. $q^*(s)$ is then pinned down by the following FOCs $f(q^*(s), s) = \mathbf{0}$, where

$$f(q, s) := \left(\frac{\sum_{i \in N_g(\beta_{*g})} [\beta_{ig}(s_{*g}) / (u_i(q, s_{i*}) - d_{ig}(q_{-g}, s))] \partial u_i(q, s_{i*}) / \partial q_g}{\sum_{i \in N_g(\beta_{*g})} \beta_{ig}(s_{*g}) / (u_i(q, s_{i*}) - d_{ig}(q_{-g}, s))} \right)_{g \in M}.$$

Denote the Jacobian of $f(q, s)$ (with respect to q) by $J(q, s)$. An interior, regular equilibrium is then defined as follows.

Definition 17. An equilibrium q^* is called interior and regular if (i) $q^* \gg \mathbf{0}$ and $Q^* < \bar{Q}$, (ii) for every firm $j \in M$, $d_{N_f(\beta_{*f})f}(q_{-f}^*, s) \ll u_{N_f(\beta_{*f})}(q_f^*, q_{-f}^*, s)$, and (iii) $J(q^*, s)$ is negative definite.

It is a maintained assumption that the equilibrium is interior and regular. Proposition 9 derives the equilibrium markup of each firm and the relationship between the weighted average Lerner index and the MHHI.

Proposition 9. In equilibrium for every firm $j \in M$ it holds that

$$\frac{P(Q^*) - C'_f(q_f^*)}{P(Q^*)} = \frac{w_f^* + \bar{\lambda}_f^*(s)}{\eta(Q^*)}.$$

The weighted average Lerner Index is $\bar{\text{LI}}(q^*) = \text{MHHI}(q^*, s) / \eta(Q^*)$.

G.4 Competitive effects of changes in corporate control

Consider an exogenous change in a shareholder's control power over a firm.

Definition 18. An exogenous increase (resp. decrease) in shareholder i 's control over firm f at $s \in S \times \mathbb{R}_+^m$ is a change in the strategic plan of firm f so that $\beta_{if}(s_{*f})$ changes infinitesimally by $d\beta_{if} > (\text{resp. } <) 0$ with all else kept constant.³⁰

Proposition 10 then studies the effects of a change in a shareholder's control over a firm.

Proposition 10. An exogenous increase (resp. decrease) in shareholder i 's control over firm f causes firm f 's quantity to change in the direction (resp. direction opposite to the one) preferred by shareholder i , that is

$$\text{sgn} \left\{ \frac{dq_f^*}{d\beta_{if}} \right\} = \text{sgn} \left\{ \frac{\partial u_i(q, s_{i*})}{\partial q_f} \bigg|_{q=q^*} \right\} = \text{sgn} \{ \bar{\lambda}_f^*(s) - \bar{\lambda}_{i,f}^*(s) \}.$$

Proposition 10 shows that if a firm is underproducing (resp. overproducing) relative to a shareholder's preferences and that shareholder's control over that firm increases, then the firm's quantity will increase (resp. decrease). The proposition also provides an intuitive measure of whether the firm is under- or overproducing relative to the shareholder's preferences. It underproduces (resp. overproduces) if its (local) weighted average Edgeworth coefficient $\bar{\lambda}_f^*(s)$ is higher (resp. lower) than the shareholder's weighted Edgeworth coefficient.

A policy proposal by Posner et al. (2017) is to require institutional investors to be passive if they accumulate large amounts of stock in multiple competing firms. Such a policy can be understood as setting $\beta_{if} = 0$ for an investment fund i and every firm f . Provided that total quantity changes in the same direction as firm f 's quantity, this policy will indeed increase consumer welfare if $\bar{\lambda}_{i,f}^*(s) > \bar{\lambda}_f^*(s)$ along a path where β_{if} 's go to 0 for every firm f .

Under WAPP, the total quantity changes in the same direction as firm f 's quantity if the game is aggregative and the slope of each firm's best response function is higher than -1 (see, e.g., Farrell and Shapiro, 1990; Vives, 1999). The game is aggregative if s is such that for every firm $j \in M$, $\lambda_{fg}(s) = \lambda_{fh}(s)$ for every pair of firms $g, h \in M \setminus \{f\}$.

H Proofs of supplementary results

Proof of Proposition 8. The game can be seen as a generalized game where the strategy constraint correspondence is $B_f^P(a_{-f}, s)$. The proof is composed of three steps.

³⁰For the entries of β_{*f} to still sum up to 1, the other entries need to decrease. However, this is just a normalization that does not affect the analysis, so it is ignored. Also, notice that an exogenous increase (resp. decrease) in d_{if} will have the same qualitative effect as an increase (resp. decrease) in β_{if} .

Step 1: $B_f^P(a_{-f}, s)$ is

- (i) nonempty: by the Remark in section 2.2, $B_f(a_{-f}, s; \pi) \neq \emptyset$; taking $\alpha_f \in B_f(a_{-f}, s; \pi)$ and $\bar{a}_f := \sum_k \alpha_f(a_f^k) a_f^k$, which lies in A_f by (i) as a finite convex combination of elements of A_f , assumption (v) and Jensen's inequality give $u_i(\bar{a}_f, a_{-f}, s_{i*}) \geq u_i(\alpha_f, a_{-f}, s_{i*}) \geq d_{if}(a_{-f}, s)$ for every $i \in N_f(\beta_{*f}(s_{*f}))$, so $\bar{a}_f \in B_f^P(a_{-f}, s)$,
- (ii) compact as a closed subset of a compact set (since u is continuous in a_f),
- (iii) upper hemicontinuous in a_{-f} , as a correspondence with closed graph to a compact space (see, e.g., Corollary 9 in p.111, Aubin and Ekeland, 1984),
- (iv) lower hemicontinuous in a_{-f} by assumption.

Also, the Nash product is continuous in a_f and a_{-f} given that u and d_{*f} are. It follows then by Berge's maximum theorem that $R_f(a_{-f}, s)$ is an upper hemicontinuous, nonempty-valued and compact-valued correspondence.

Step 2: For any $i \in N_f(\beta_{*f}(s_{*f}))$ and any $a_{-f} \in A_{-f}$ we have that $u_i(a_f, a_{-f}, s_{i*}) - d_{if}(a_{-f}, s)$ is concave in a_f over $B_f^P(a_{-f}, s)$. It follows that for any $i \in N_f(\beta_{*f}(s_{*f}))$ and any a_{-f}

$$(u_i(a_f, a_{-f}, s_{i*}) - d_{if}(a_{-f}, s))^{\beta_{if}(s_{*f})}$$

is concave (and thus log-concave) over $B_f^P(a_{-f}, s)$, since $a_f \mapsto u_i(a_f, a_{-f}, s_{i*}) - d_{if}(a_{-f}, s)$ is concave and $x \mapsto x^{\beta_{if}(s_{*f})}$ is concave and increasing. Thus,

$$\prod_{i \in N_f(\beta_{*f}(s_{*f}))} (u_i(a_f, a_{-f}, s_{i*}) - d_{if}(a_{-f}, s))^{\beta_{if}(s_{*f})}$$

is log-concave over $B_f^P(a_{-f}, s)$ as a product of log-concave functions (and thus also quasiconcave in a_f for every a_{-f}). The product is also continuous in a_f and a_{-f} , and given also that $B_f^P(a_{-f}, s)$ is convex for any $a_{-f} \in A_{-f}$ as an intersection of superlevel sets of functions that are concave over the convex set A_f , it follows that $R_f(a_{-f}, s)$ is convex-valued.

Step 3: $G(a) := \times_{f \in M} R_f(a_{-f}, s)$ is an upper hemicontinuous, nonempty-, compact- and convex-valued correspondence since R_f is for each $f \in M$. By Kakutani's fixed point theorem, G admits a fixed point, which is an equilibrium. **Q.E.D.**

Proof of Lemma 3. The derivative of $u_i(q_f, q_{-f}, s_{i*})$ with respect to q_f is given by

$$\frac{\partial u_i(q_f, q_{-f}, s_{i*})}{\partial q_f} = s_{if} \left(P(Q) - C'_f(q_f) \right) + P'(Q) \sum_{g \in M} s_{ig} q_g,$$

and the second derivative by

$$\begin{aligned}\frac{\partial^2 u_i(q_f, q_{-f}, s_{i*})}{\partial q_f^2} &= 2s_{if}P'(Q) - s_{if}C_f''(q_f) + P''(Q) \sum_{g \in M} s_{ig}q_g \\ &= P'(Q) \left[s_{if} \left(2 - \frac{C_f''(q_f)}{P'(Q)} \right) - E(Q) \sum_{g \in M} s_{ig}w_g \right],\end{aligned}$$

and the result follows. **Q.E.D.**

Proof of Lemma 4. Since for $q_f > \bar{q}$ profit becomes negative, we can constrain each firm to choose quantity $q_f \in [0, \bar{q}]$. From continuity of u_i in q_f and the definition of B_f^P it follows then that B_f^P is compact. Especially, given strict concavity of u_i in q_f for every i , it follows that B_f^P is convex, thus a closed interval. We distinguish the following two cases:

Case 1: Given that u_i is strictly concave in q_f for every i (so u_i can be equal to d_{if} for at most 2 values of q_f in B_f^P), the only way that $\forall q_f \in B_f^P$ there exists $i \in N_f(\beta_{*f}(s_{*f}))$ such that $d_{if}(q_{-f}, s) = u_i(q_f, q_{-f}, s_{i*})$ is for B_f^P to be a singleton. Part (i) then holds, part (ii) follows since $R_f(q_{-f}, s) = B_f^P$, and part (iii) holds because B_f^P contains no q_f other than $R_f(q_{-f}, s)$.

Case 2: If $\exists q_f \in B_f^P$ such that $u_i(q_f, q_{-f}, s) > d_{if}(q_{-f}, s)$ for every $i \in N_f(\beta_{*f}(s_{*f}))$, we have that for every $i \in N_f(\beta_{*f}(s_{*f}))$ and every $q_f \in B_f^P(q_{-f}, s)$

$$\begin{aligned}&\frac{\partial^2 (u_i(q_f, q_{-f}, s_{i*}) - d_{if}(q_{-f}, s))^{\beta_{if}(s_{*f})}}{\partial q_f^2} \\ &= - \frac{\beta_{if}(s_{*f})(1 - \beta_{if}(s_{*f}))}{(u_i(q_f, q_{-f}, s_{i*}) - d_{if}(q_{-f}, s))^{2 - \beta_{if}(s_{*f})}} \left(\frac{\partial u_i(q_f, q_{-f}, s_{i*})}{\partial q_f} \right)^2 \\ &\quad + \frac{\beta_{if}(s_{*f})}{(u_i(q_f, q_{-f}, s_{i*}) - d_{if}(q_{-f}, s))^{1 - \beta_{if}(s_{*f})}} \frac{\partial^2 u_i(q_f, q_{-f}, s_{i*})}{\partial q_f^2} < 0,\end{aligned}$$

by strict concavity of u_i in q_f . Also, for every $i \in N_f(\beta_{*f}(s_{*f}))$, $(u_i(q_f, q_{-f}, s_{i*}) - d_{if}(q_{-f}, s))^{\beta_{if}(s_{*f})}$ is non-negative and not identically equal to zero over B_f^P . The results then follow from Theorem 4 in Kantrowitz and Neumann (2005). **Q.E.D.**

Proof of Proposition 9. The FOCs in equilibrium give:

$$P(Q^*) - C_f'(q_f^*) + P'(Q^*) \left[q_f^* + \sum_{g \in M \setminus \{f\}} \lambda_{fg}^*(s) q_g^* \right] = 0,$$

and the result follows. **Q.E.D.**

Proof of Proposition 10. The partial derivative of $f(q, s)$ with respect to β_{if} is

$$\begin{aligned} \frac{\partial f(q, s)}{\partial \beta_{if}} &= \left[\frac{\gamma_{if}^*}{\beta_{if}} \frac{\partial u_i(q, s_{i*})}{\partial q_f} - \frac{1}{u_i - d_{if}} \frac{1}{\sum_{j \in N_f(\beta_{*f}(s_{*f}))} \frac{\beta_{jf}}{u_j - d_{jf}}} \sum_{k \in N_f(\beta_{*f}(s_{*f}))} \gamma_{kf}^* \frac{\partial u_k(q, s_{k*})}{\partial q_f} \right] \cdot \mathbf{e}_f \\ &= \frac{\gamma_{if}^*}{\beta_{if}} \frac{\partial u_i(q, s_{i*})}{\partial q_f} \cdot \mathbf{e}_f, \end{aligned}$$

where $\mathbf{e}_f$ the m -dimensional standard unit vector with 1 in its f -th dimension. It follows by the Implicit Function Theorem that

$$\begin{aligned} \begin{pmatrix} \frac{dq_1^*}{d\beta_{if}} \\ \frac{dq_2^*}{d\beta_{if}} \\ \vdots \\ \frac{dq_m^*}{d\beta_{if}} \end{pmatrix} &= -J^{-1}(q^*, s) \left. \frac{\partial u_i(q, s_{i*})}{\partial q_f} \right|_{q=q^*} \cdot \mathbf{e}_f = -(\det(J))^{-1} \frac{\partial u_i}{\partial q_f} \cdot \text{adj}(J) \mathbf{e}_f \\ &= -(\det(J))^{-1} \frac{\partial u_i}{\partial q_f} \cdot \begin{pmatrix} (-1)^{1+f} \det(J_{-f-1}) \\ (-1)^{2+f} \det(J_{-f-2}) \\ \vdots \\ (-1)^{m+f} \det(J_{-f-m}) \end{pmatrix}, \end{aligned}$$

where the second equality follows from the Laplace expansion, $\text{adj}(J)$ is the adjugate or classical adjoint of J , and J_{-f-g} is the J matrix with the f -th row and g -th column removed. Since J is negative definite

$$\begin{aligned} \text{sgn}\{\det(J)\} &= -\text{sgn}\{\det(J_{-f-f})\} = \text{sgn}\{(-1)^m\}, \\ \text{so that } \text{sgn}\left\{\frac{dq_f^*}{d\beta_{if}}\right\} &= \text{sgn}\left\{(-1)^{2f} \frac{\partial u_i}{\partial q_f}\right\} = \text{sgn}\left\{\left. \frac{\partial u_i(q, s_{i*})}{\partial q_f} \right|_{q=q^*}\right\}, \end{aligned}$$

where

$$\begin{aligned} \left. \frac{\partial u_i(q, s_{i*})}{\partial q_f} \right|_{q=q^*} &= \sum_{g=1}^m s_{ig} \left. \frac{\partial \pi_g(q, s_{i*})}{\partial q_f} \right|_{q=q^*} = P(Q^*) \left[s_{if} \frac{P(Q^*) - C'_f(q_f^*)}{P(Q^*)} - \frac{\sum_{g=1}^m s_{ig} w_g^*}{\eta(Q^*)} \right] \\ &= -Q^* P'(Q^*) \left[s_{if} (w_f^* + \bar{\lambda}_f) - \sum_{g=1}^m s_{ig} w_g^* \right] = -Q^* P'(Q^*) s_{if} (\bar{\lambda}_f^* - \bar{\lambda}_{i,f}^*), \end{aligned}$$

and the result follows. **Q.E.D.**

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